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A review on polymeric insulation for high-voltage application under various stress conditions

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Abstract

Polymeric material draws a great attraction in the development of insulation for High Voltage Direct Current (HVDC) applications. Insulation material is the deciding key factor to increase the operating voltage and to transmit the bulk power with minimized power loss. Premature breakdown, high withstanding temperature, and impacts of space charge accumulation are all issues faced by the insulation material. Material qualities such as dielectric, electrical, structural, and mechanical properties are directly affected by the issues. Therefore, this paper provides an overview of the dielectric properties of polymeric materials and their composites under various stress conditions. The stress includes AC, DC, superimpose and combined (AC and DC) stress under which the dielectric constant, dielectric losses, electrical strength, and space charge distribution of polymeric insulating materials such as low-density polyethylene, high-density polyethylene, cross-linked polyethylene, polypropylene, and random copolymer (RCP), as well as their composites, are explored. Polypropylene and RCP outperform all other polymers in terms of breakdown strength and space charge suppression under various stress conditions. RCP is a developing material that also has the potential to be an effective thermoplastic insulation material in the future.

KEYWORDS

electrical property, nanofillers, polypropylene, random copolymer, thermoplastic

1 | INTRODUCTION

(Globally, power consumption is increasing every day; hence, most research is currently focused on renewable energy sources to meet this demand.^[1-3] Among the renewable energy sources, wind energy contributes its major share worldwide.^[4-6] To transmit the power from the generator to the load center with minimized power loss is the big deal. Hence, there is continuous development in the power cable on minimizing loss as the bulk power is transmitted in both HVAC and HVDC applications. When compared to High Voltage Alternate current (HVAC) transmission, HVDC transmission is more

suitable for transmitting large amounts of power over long distance. Due to the absence of reactive power losses, conductor size is reduced and it does not require any additional compensator circuit. HVAC transmission is not preferred for submarine or underwater power transmission due to stray capacitance of the power cable and hence, HVDC transmission is becoming more prominent for transmitting power from on- and off-shore wind farms.^[4,5,7,8]

Oil-filled cables, also known as fluid-filled cables, were initially used,^[7] in which the main insulation consists of multi-layer impregnated Kraft papers. Oil-filled (OF) cables are classified as Low viscosity oil and high

viscosity oil, the former is utilized for a maximum length of 30–60 km, and the latter does not have any limitation in its length. Polypropylene laminated paper is used as an insulation matrix in OF cable, which improves its performance. Although OF cable is operated at high voltage, it has limitations such as cable length and the threat of oil leakage.^[9–11] As a result, the OF cable has been replaced by the mass impregnated (MI) cable. The principal insulation of the mass-impregnated cable is likewise Kraft paper, comparable to that of OF cable^[9,12] but has low dielectric strength.

To achieve greater dielectric characteristics, high-density sheets are commonly utilized. This high-density MIHVDC cable has shown excellent performance in HVDC applications for more than 60 years.^[13] However, in the 1940s, Polymer-based insulation became popular and extensively used in HVDC applications because of its low dielectric loss, higher breakdown strength, and excellent mechanical and thermal properties. For cable applications, polymeric insulating materials like polyethylene, ethylene-propylene diene monomer rubber, and polypropylene (iPP) is used.^[14] In common, Polyethylene is utilized as an insulating material due to its high breakdown strength and low permittivity, making it a better choice for HV insulation. However, it has several drawbacks such as low thermal conductivity hence it cannot be operated at high temperatures and poor mechanical properties.

Cross-linked polyethylene (XLPE) has been developed to overcome the drawbacks of polyethylene. XLPE is a conventional insulation material that addresses the problems outlined above.^[15,16] XLPE is being used as an insulating material in both HVAC and HVDC cables.^[17–20] However, during cross-linking, the by product will cause a significant accumulation of space charges, which will degrade the cable's insulation and cause premature failure.^[21–24] Moreover, XLPE is a thermoset material; it cannot be recycled or reused at the end of its useful life.^[25] Limitation in cross-linking in manufacturing on XLPE cables and environmental concerns causes more investigation on thermoplastic materials. Polypropylene (iPP) is a thermoplastic with superior electrical, mechanical, and thermal qualities.^[26,27] In addition to that, iPP has the advantage of having a higher breakdown strength and a high withstanding temperature. At room temperature, it suffers from brittleness and stiffness. However, by combining or copolymerizing with additional monomers, this feature can be changed. iPP blending with other materials gives an excellent dielectric profile but incompatibility and phase separation under long-term operation are still an issue. Apart from blending, copolymerization with another polymer improves the polymer properties and avoids the problem associated with the incompatibility and phase separation.

In recent, there is limited research dealing with polypropylene copolymer. In comparison with other polymer composites, random copolymer (RCP) in virgin exhibits higher breakdown strength and observed significant space charge suppression. The research work on the RCP composites is very scanty. The creation of novel thermoplastic materials and nanocomposites has been the focus of the current research effort. By incorporating nanosized organic and inorganic particles into the polymer material, the electrical and mechanical properties of the polymer can be fine-tuned. With varying filler loadings in the polymer, the nanocomposite performs better than the virgin material. The interaction of the fillers and polymers has an impact on the chemical and physical properties.^[28–30]

During long-distance power transmission, the electronic circuits involved stress the cable insulation significantly. Therefore, to employ environment-friendly dielectric materials, materials such as RCP needs to be explored. To develop the insulation to withstand such undue stress, the dielectric materials need to be studied with stress conditions such as AC, DC, and combined (AC + DC) stress. This work reviews the dielectric constant, space charge, breakdown strength, and conductivity of various polymeric dielectric materials including the environment-friendly ones and their composites stressed under AC, DC, impulse, and combined (AC and DC) stress for high-voltage applications. The following materials are covered: Low-density polyethylene (LDPE), high-density polyethylene (HDPE), XLPE, iPP, and RCP.

2 | CLASSIFICATION OF POLYMERIC INSULATION CABLES

The majority of power generation occurs far from the load center, hence it is necessary to transmit power over a long distance with minimized losses. Converters are used at both ends of the cable transmission for conversion making the power transmitted with distorted voltage and current waveform. As a result, the insulation will get stressed under combined stress (DC voltage with harmonics). Similarly, during transmission, the dielectric properties of the insulation may get altered according to different manufacturing conditions. The properties of the insulation should be examined under various stresses to reveal the behavior of the insulation. Researchers have reported on the insulation behavior under various stress conditions, which is explained in detail in this section. Previously, HDPE and LDPE have been widely used insulation materials, but due to poor mechanical properties and low withstanding temperature, it is replaced by XLPE. XLPE is more widely used than LDPE and HDPE because of its superior thermomechanical properties.

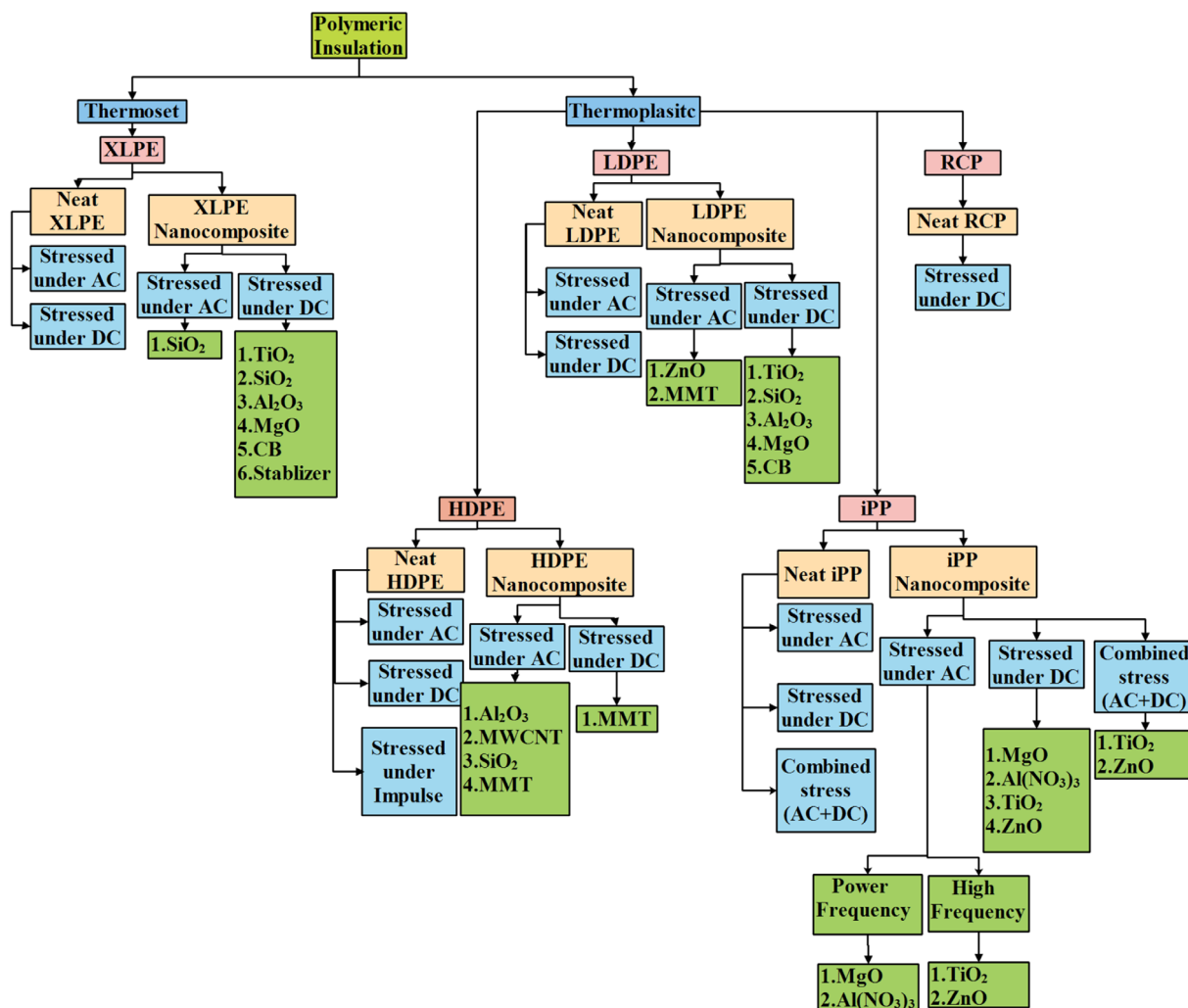


FIGURE 1 Classification of polymeric insulation.

Even though XLPE has superior characteristics to LDPE but it cannot be recycled due to its thermosetting nature. So the researchers addressed the issues faced by the XLPE by replacing them with thermoplastic material. iPP and RCP, on the other hand, are being used in new insulation materials. Both have a high breakdown strength and minimize the accumulation of space charges. The broad classification of the polymeric cable is shown in Figure 1.

2.1 | High-density polyethylene

2.1.1 | Neat HDPE

Researchers developed a new polymeric material derived from polyethylene called HDPE to address the drawbacks of MI cables, such as low-temperature stability and mechanical properties. HDPE has a carbon-carbon backbone. It is created by polymerizing an ethylene monomer

in the presence of a catalyst. HDPE has little branching and has a large spherulite size, strong crystallinity, and dielectric constant fluctuation with frequency.

2.1.2 | Modified HDPE (m-HDPE) under AC stress

According to reports from various researchers, modifying the HDPE material under various manufacturing conditions improves the dielectric strength and suppresses the material's space charge distribution.^[31–33] Lessard et al., reported on the effect of temperature on the breakdown behavior of low and HDPE at temperature ranges from 23 to 160°C. The sample was subjected to a voltage ramp of 3 kV/s at 60 Hz. According to the statistics, the breakdown strength appears to decrease as the temperature rises. The breakdown value of both polymers decreases from around 8 MV/cm at 25°C to 3 MV/cm at melting temperature. The decrease in the breakdown strength of

the dielectric material depends on the crystallization behavior of the polymer and it is inversely proportional to the inverse of the free volume cavity.^[34] Shah et al., investigated the characteristic performance of the clay nanocomposite of HDPE. From the observation, the virgin HDPE under the voltage stress rate of 500 V/s shows a breakdown strength of 30 kV/mm and its resistivity is 2.6×10^{12} . Ramkumar et al.,^[35] also reported a similar observation for neat HDPE. Praveen Kumar et al., reported that the breakdown strength of the neat HDPE is 23 kV/mm at the voltage rate of 1 kV/s. The authors have also dealt with the dielectric properties such as resistivity and permittivity with the nonpolar HDPE. The resistivity of the HDPE is reported the same as the breakdown profile. Niwia et al., also reported that the dielectric properties of HDPE are altered by its manufacturing parameter. The manufacturing parameter and its annealing of the sample exhibit higher breakdown strength than the previously reported and its ranges from 27 to 50 kV/mm. The breakdown strength of the HDPE depends on the crystalline changes in the HDPE under the annealing process. Dabbak et al., reported on a comparative study of the dielectric strength of the HDPE and PP. From the observation, HDPE exhibits higher breakdown strength than the PP. The breakdown strength of the HDPE is 70 kV/mm whereas for PP it is 58 kV/mm when stressed at the rate of rise of 500 V/s. According to the observation, the dielectric strength of HDPE is 30 kV, higher than that of pure PP, and 7 kV lower than that of LDPE. The degree of crystallinity and the formation of crystalline material have a significant impact on the breakdown strength. HDPE has a lower melting temperature and a lower crystallization peak temperature than PP, altering the crystal structure.^[14]

2.1.3 | Modified HDPE under DC stress

The influence of crystallinity on the breakdown strength and conduction property of the polyethylene polymer under dc stress is reported. According to the findings, the polymer's breakdown strength increases as crystallinity increases, while the free volume path shrinks. As a result, the free-electron path and the electron accelerating under the field are both shortened, resulting in increased breakdown strength. The breakdown strength of HDPE, MDPE, and LLDPE is 37.96%, 28.56%, and 15.01% higher than that of LDPE.^[36] Suzuki et al., reported dielectric properties of HDPE under DC stress. The dielectric strength of the HDPE is altered by the inclusion of an ionomer. According to the findings, as density increases, HDPE has a higher breakdown strength than LDPE, and modified HDPE has a higher breakdown strength than unmodified HDPE as well as other polymer materials. The modified HDPE shows higher breakdown strength than the unmodified

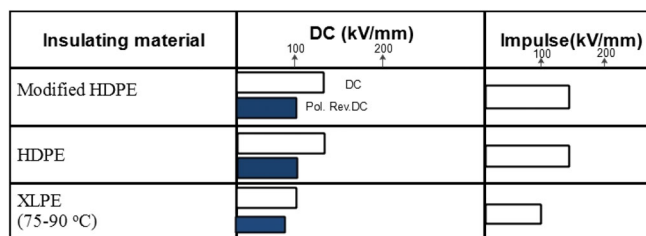


FIGURE 2 Breakdown strength of the HDPE and modified one comparing with XLPE.^[37]

one and it is in the range of 410 kV/mm and its resistivity also improved. Modified HDPE has an excellent breakdown strength due to its high crystallinity and low space charge accumulation. According to the research, the modified HDPE has a breakdown strength twice that of XLPE, as shown in the redrawn Figure 2.^[36,37] Thoru et al., reported on the dielectric strength of the various polymeric material. Among these HDPE shows higher breakdown strength and its 400 kV/mm. The breakdown strength of the polymeric material is influenced by the degree of crystallization and it is also already reported by the Lizzard.

2.1.4 | Modified HDPE under impulse stress

Niwa et al., reported on the dielectric properties of the HDPE under impulse stress. The dielectric strength under impulse voltage is influenced by the crystallization under annealing processes and the degree of crystallinity structures of the crystallization alter the breakdown strength. By varying annealing temperature, the breakdown strength increased from 140 to 240 kV/mm. Under certain manufacturing conditions, the breakdown value of HDPE is 1.6 times that of XLPE.^[38] Thoru Takahashi et al., also reported on the impulse breakdown strength of the HDPE, from the observation the breakdown strength is influenced by the crystallization of the polymer. The crystallization of the polymer is influenced by the nucleating agent and cooling speed. The nucleating agent regulates the size of the spherulite and accelerates crystallization, while the cooling rate optimizes the conditions for maximum crystal growth. At optimum production conditions, 60% of HDPE cable has a higher breakdown strength than conventional XLPE cable.^[39]

2.1.5 | Neat HDPE - space charge

Daisuke Muto reported on the space charge properties of modified HDPE under various electrode and temperature conditions. Even at low fields, positive homo charge

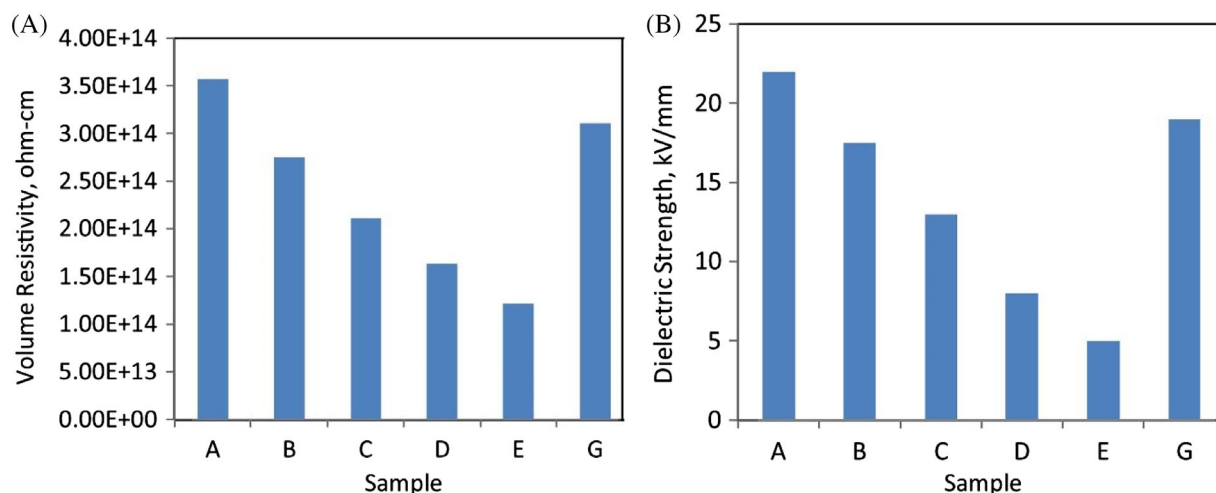


FIGURE 3 HDPE nanocomposite (A) resistivity and (B) breakdown strength.^[41]

dominates in m-HDPE with Au electrodes, while negative charges form at 80°C. As the temperature increased, the negative charge injection became stronger. The current under the Semicon layer was higher than in m-HDPE with Au electrodes. This could be a result of contaminants diffusing through the Semicon layer during heat treatment.^[38]

2.1.6 | HDPE nanocomposite under AC stress

It has been reported that the addition of multi-wall carbon nanotubes and nano-SiO₂ at various weight percentages (1, 2, 3, 4, and 5 wt%) improves the electric and mechanical properties of HDPE. According to the observations, the volume of resistivity follows the same trend as the breakdown strength. Due to high bandgap energy and atomic configuration, SiO₂ nanocomposite exhibits higher breakdown strength under AC stress than neat HDPE. The volume resistivity of nano-SiO₂ 5 wt% is 3.11×10^{-14} V-cm, which is close to the values of pure HDPE, and the dielectric strength is maximum when compared to neat and composite with MWCNT. When 4 wt% SiO₂ and 1 wt% MWCNT are combined, the polymer's tensile strength increases by 82%; however, at higher loading, the tensile strength begins to decrease. The breakdown strength and resistivity of the HDPE nanocomposite are depicted in Figure 3, which shows that SiO₂ nanoparticles have a higher breakdown strength than MWCNT, and the higher breakdown strength is dependent on higher SiO₂ content and low MWCNT filling concentration.^[41]

Ramkumar et al., investigated the dielectric properties of HDPE with an alumina nanofiller. By incorporating of nanofiller the breakdown strength of the nanocomposite is increased with an increase in filler concentration up to

3 wt% and it is 20.8% greater than the neat LDPE. The capacitance of the material increases as the dielectric constant increases, and it is tending to reduce dissipation loss. The dissipation loss of the composite material at 7 wt% is small when compared to other filler concentrations. Hence, the dielectric factor influences the material's AC breakdown strength, the breakdown value at 5 wt% is 16.33% higher than pure HDPE. Similarly, the breakdown value of Al₂O₃/HDPE composite is higher than that of SiO₂ nanocomposite. The dielectric constant rises as the filler concentration rise and the loss factor fall until a certain weight percentage is reached, at which point it begins to rise again.^[35] According to Gouda et al., the inclusion of clay nanofiller improved the properties of HDPE. Hexadecyl trimethyl amino bromide is used to modify Na-MMT clay. With increasing filler concentration, the dielectric strength increases and then begins to decrease. The optimal value of break down strength in the filler concentration is 14.4 MV/m at 6 wt%, but it decreases at 10 and 15 wt%. However, at a filler concentration of 10 wt%, the breakdown strength exceeds that of pure HDPE. The charges trapped at the polymer-nanoparticle interface cause the composite material's higher breakdown value. The breakdown strength at 6 wt% is 15% greater than that of neat polymer. Additionally, the presence of filler lowers the dissipation factor. The loss factor decreases up to 6 wt% and then begins to increase above 10 wt% as filler concentration increases.^[42]

2.1.7 | HDPE nanocomposite under DC stress

Shah et al., reported that the addition of montmorillonite clay to HDPE improves both its electrical and mechanical

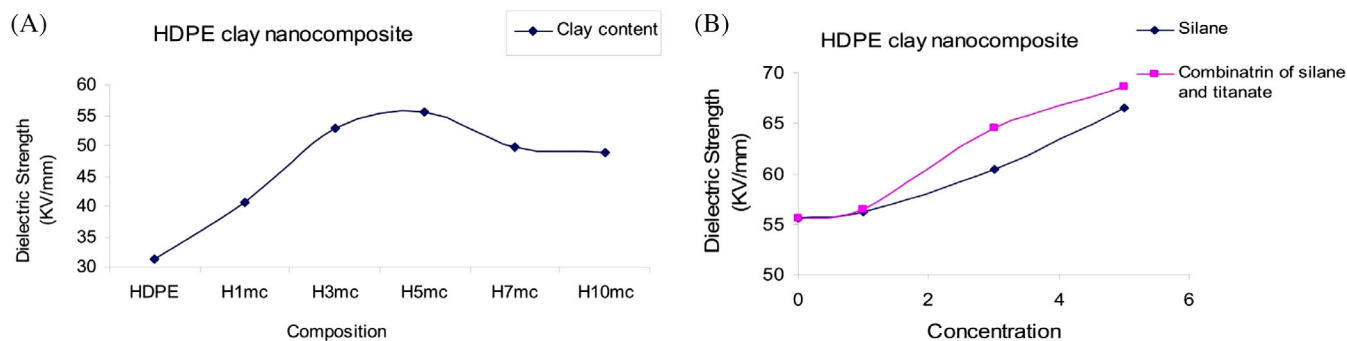


FIGURE 4 Dielectric strength of (A) HDPE nanocomposite and (B) HDPE nanocomposite treated with silane and titanate.^[38]

properties under DC stress. To make it organophilic, indigenous montmorillonite clay is first treated with hexadecyl trimethyl ammonium bromide, then surface treated with aminopropyl triethoxysilane, and finally with titanium acetylacetonate. The composite sample is composed of varying percentages of HnmC (H refers to HDPE and n is the weight percentage of organoclay) and 5 wt% of organoclay treated with varying percentages of silane and titanate coupling agent. The test specimen is stressed under DC voltage at the rate of 500 V/s. With a 5% increase in filler concentration, the dielectric, and tensile strengths of the nanocomposite increase dramatically.

The unmodified and modified clay composites have breakdown values of 48 and 54 kV/mm, respectively, whereas the neat HDPE has a breakdown value of 30 kV/mm, as shown in Figure 4. It is already clear that under AC stress, the breakdown strength of the clay nanocomposite is increased by 15%.^[41] The clay's exfoliated and intercalated structure acts as a barrier, hence it increasing the path for the breakdown.^[42] Table 1 shows the overview of HDPE.

The study of HDPE and its composites in the literature under various stresses are summarized as:

- The breakdown strength of pure HDPE under AC stress is fairly low. The modified HDPE has low crystallinity and it possesses higher breakdown strength. Similarly, the accumulation of space charge is also low.
- Under DC stress, HDPE exhibits slightly higher dielectric strength than AC, and also the breakdown strength of the material can be increased with the increasing thickness of the sample.
- At 5% weight, clay nanofiller improves the AC and DC breakdown strength and tensile properties of HDPE.
- Under AC stress, the addition of Al_2O_3 at 5 wt% results in a 16% higher breakdown strength than pure HDPE, as well as a 20.18% increase in dielectric constant.
- The addition of nano- SiO_2 filler increases the breakdown strength of the pure HDPE and also, increases the volume of resistivity, which is close to that of pure HDPE.

2.2 | Low-density polyethylene

2.2.1 | LDPE neat

LDPE has a saturated carbon-carbon backbone structure and is derived from an ethylene monomer. LDPE is better suited for high-voltage cable insulation because of its low dielectric loss, low dielectric constant, and high breakdown strength.^[43] In a DC cable, space charge accumulation causes local electric field distortion and damage the cable. The applied electric field and temperature have an impact on the material's space charge distribution.

2.2.2 | Neat LDPE under AC stress

Fauqiang Tian et al.,^[45] reported on the electrical properties of LDPE for the test specimen with a thickness of 100–200 μm stressed at a rate of 1 kV/s under AC voltage. From the observation, LDPE exhibits low conduction current value, and the breakdown value of virgin LDPE is 90 kV/mm. Ruijin et al.,^[46] also reported on properties of LDPE, when stressed under alternating current-voltage (50 Hz) at a rate of 500 V/s and the breakdown value of pure LDPE is 97.9 kV/mm. Furthermore, the resistivity of the sample is $2.4 \times 10^{17} (\Omega \text{ cm})$. The reason behind the dielectric breakdown strength of the material is due to the density of the free electrons and the energy that they gain, hence this will lead to a decrease in the electrical breakdown strength.

2.2.3 | Neat LDPE DC stress

Si-Jiao Wang et al.,^[47] reported on the properties of LDPE. A thin-film sample with a thickness of 30 μm is subjected to a DC voltage at a rate of 500 V/s and the breakdown value of the neat material is 464 kV/mm. According to the findings, the pure material's permittivity and dielectric loss increase as frequency decreases. Similarly, Jun-Guo-Gao et al.,^[48] also reported on the

TABLE 1 HDPE overview

Material	Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity (Ω cm)	Electric field (kV/mm)
HDPE	Neat	AC	—	Plane-Plane	—	—	2.3	No data	10 ¹⁴	64 ^[14]
			—	Cylinder-Cylinder	3000	1	2.7	0.018828	—	28 ^[35]
			—	Cylinder-Cylinder	3000	1	Decreases with increase in frequency	Decreases with increase in frequency	3.54 × 10 ¹⁴	23 ^[41]
			—	Plane-Plane	2500	0.75	High at a lower frequency	High at a lower frequency	—	12.52 ^[42]
			—	Sphere-Sphere	—	3	No data	No data	—	8 ^[34]
	Composite	DC	—	Plane-Plane	200–300	2	2.3	0.05	9 × 10 ¹⁶	330 ^[37]
			—	Cylinder-Cylinder	—	0.5	No data	No data	2.6 × 10 ¹²	31 ^[43]
			—	Sphere-Plane	200	(1 × 0.040 ms)	No data	No data	—	215 ^[38]
			—	Sphere-Plane	200–300	1 × 40 μm	No data	No data	—	350 ^[39]
			Clay	Plane-Plane	2500	0.75	Increases with increase in frequency and temperature	Increases with increase in frequency and temperature	—	14.4 ^[42]
		AC	MWCNT	Cylinder-Cylinder	3000	1	Increases with filler and decreases with frequency	Increases with filler and decreases with frequency	3 × 10 ¹⁴	20 ^[41]
			SiO ₂	Cylinder-Cylinder	3000	1	Increases with increase in filler content	Decreases with increase in filler content	—	32.79 ^[35]
			Al ₂ O ₃	Cylinder-Cylinder	3000	1	Increases with increase in filler content	Decreases with increase in filler content	—	—
			Clay	Plane-Plane	—	—	2.21	No data	10 ¹⁶	50.9 ^[14]
		DC	Clay	Cylinder-Cylinder	—	0.5	No data	No data	19.6 × 10 ¹⁵	68.7 ^[43]

properties of LDPE. The sample with a thickness of 200 μm is stressed at the rate of 1 kV/s. According to the observations, the pure material's electric conductance increases with the field, and the breakdown strength is 299 kV/mm. Wang et al.,^[49] also revealed the properties of LDPE. The dielectric property and breakdown strength of the pure LDPE sample has a thickness of 0.2 mm and is stressed at a rate of 0.5 kV/s under DC voltage. The breakdown value of pure LDPE is 223.3 kV/mm, and the permittivity of the LDPE is independent of frequency.

2.2.4 | Neat LDPE space charge

The amount of space charge accumulation increases with an increasing applied field and decreases with an increasing temperature. In addition, the space charge distribution in LDPE is said to be material-dependent.^[50–53] Nurizani Hussin et al., reported a study on space charge accumulation in LDPE by introducing cross-linking by products, which revealed that, when compared to fresh material, there is a greater amount of space charge accumulation in LDPE due to cross-linking by products.^[54,55] It is also reported that the annealing treatments influence the accumulation of space charge and also the researchers reported that space charge formation in XLPE and LDPE is influenced not only by the cross-linking agent but also by the annealing process used during production.^[56–58]

2.2.5 | LDPE nanocomposites

The addition of organic or inorganic nanoparticles improves the electrical and mechanical properties of the polymer matrix. As a result, it encourages researchers to work on various nanoparticles to improve the performance of nanocomposites.

2.2.6 | LDPE nanocomposite under AC stress

Conducting nanofiller such as carbon black nanofiller in the polymer matrix reduces space charge formation, dielectric permittivity and also reduces the loss tangent. But in contrast, increasing the filler loading lowers the material's breakdown strength. Agglomeration is possible at higher filler loadings, resulting in faster breakdown.^[59–61] Fauqiang Tian et al.,^[45] reported on the electrical properties of LDPE nanocomposites. The test specimen with a thickness of 100–200 μm is stressed at a rate of 1 kV/s under AC voltage at various weight percentages of ZnO (0, 0.1, 0.5, 1, 3, 5, and 7 wt%). According to the findings, the conduction current in

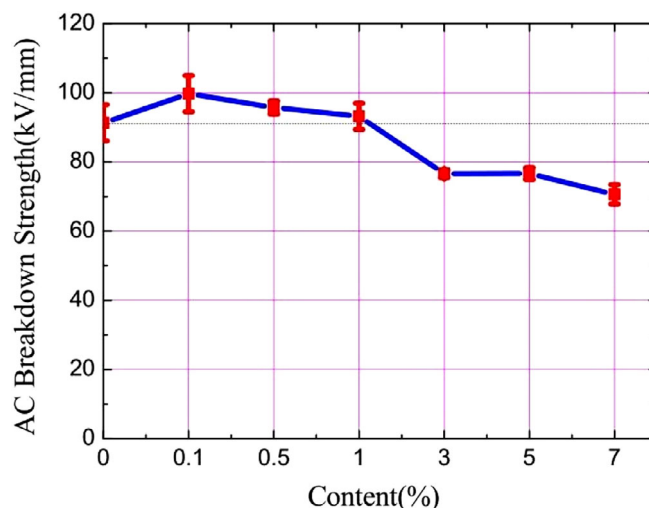


FIGURE 5 Breakdown characteristic of LDPE and its nanocomposite under AC stress.^[45]

the LDPE composite is significantly lower in both low and high fields. The polymer matrix's nanoparticles act as a nucleating agent and also accelerate crystalline speed will be preventing large spherulite formation. Hence, this will influence the breakdown strength and it has a higher breakdown value at low concentrations and begins to decrease at concentrations greater than 1 wt%, as illustrated in Figure 5. The breakdown strength is 100 kV/mm at 0.1 wt% filler loading, which is higher than the breakdown strength of neat LDPE. Ruijin Liao et al., also reported on the improvement of electrical properties by incorporating nanofillers. Montmorillonite (MMT) nanoclay is surface modified with octadecylamine and a silane coupling agent before it is incorporated into the polymer matrix.

PEM1, PEM3, and PEM5 are the names given to LDPE modified with MMT contents of 1, 3, and 5 wt%, respectively. A sample with a thickness of 100 μm is subjected to an alternating current voltage (50 Hz) at a rate of 500 V/s until it breaks down. The MMT sheet in the polymer reduces charger carrier mobility and increases resistivity. The conduction current gets reduced due to an increase in the resistivity and hence this leads to an increase in the electrical breakdown strength. It depicts that, as the filler content increases, so does the volume of resistivity and the material's breakdown strength. According to the data in Figure 6, the addition of MMT to LDPE increases its breakdown strength by 11% when compared to neat LDPE.^[46]

2.2.7 | LDPE nanocomposite under DC stress

According to Shan Luo et al., the addition of carbon black nanofiller changes the properties of the polymer

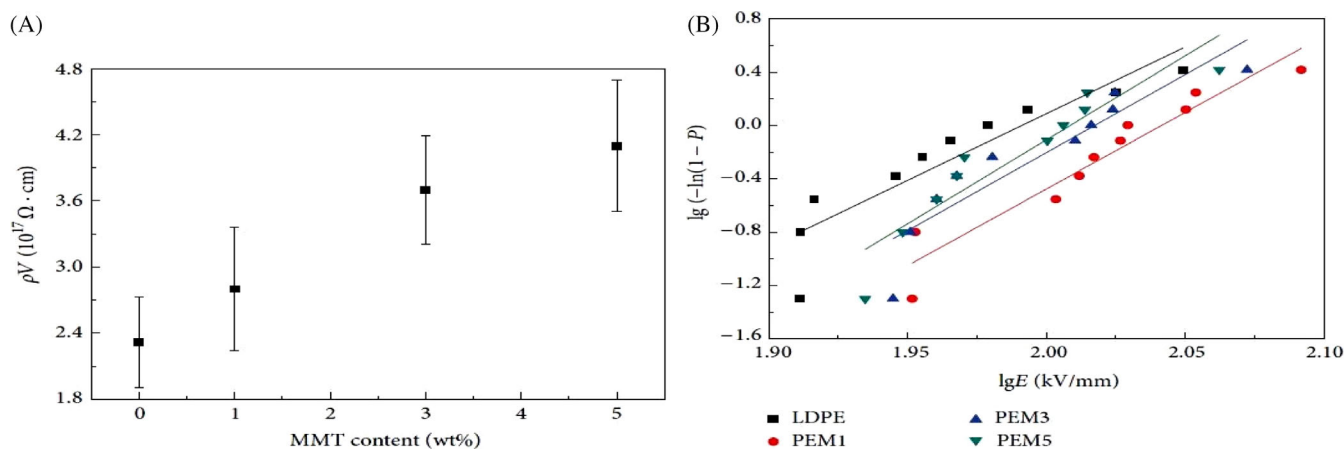


FIGURE 6 (A) Resistivity and (B) breakdown of the LDPE/MMT nanocomposite.^[46]

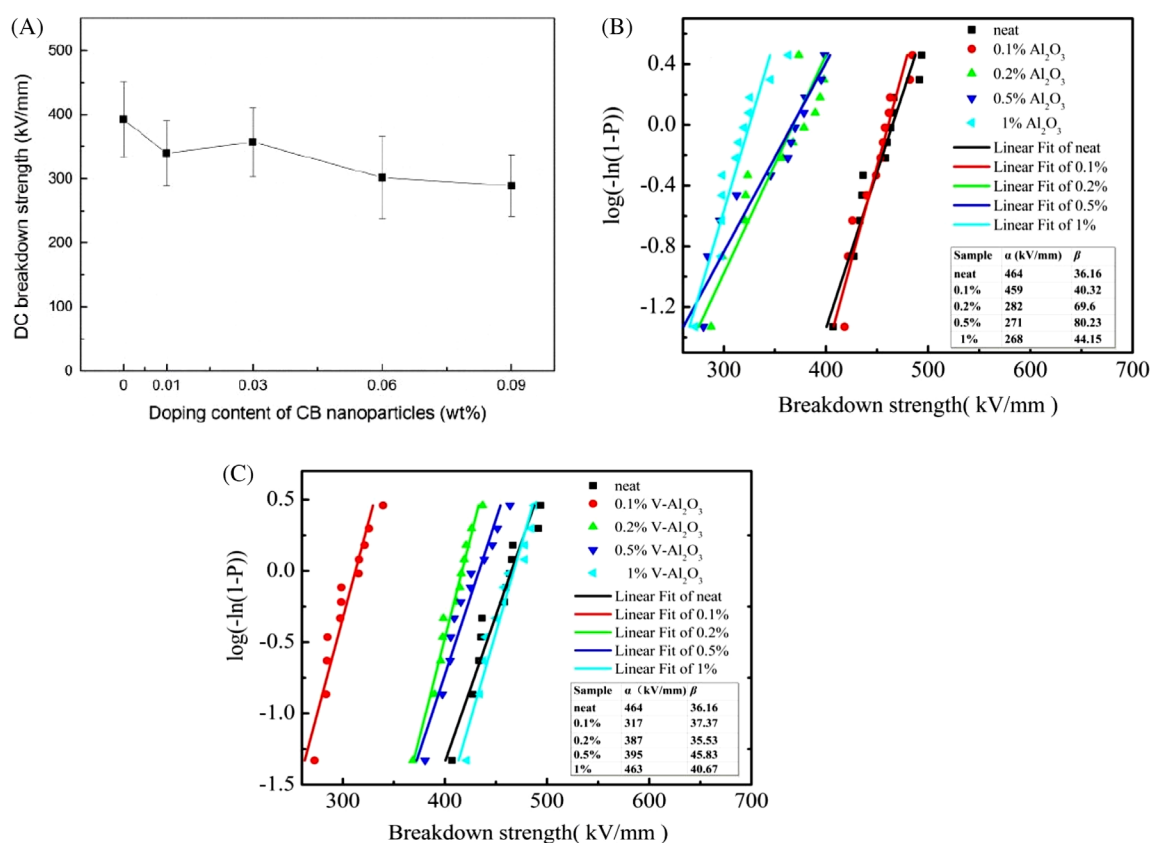


FIGURE 7 Breakdown characteristics of (A) LDPE/CB,^[60] (B) LDPE/Al₂O₃, and (C) LDPE/V-Al₂O₃ nanocomposite.^[47]

nanocomposites. The inclusion of nanoparticles in the polymer reduces the volume of resistivity, which is lower than that of neat LDPE as observed. As a result, decreasing the volume of resistivity reduces the nanocomposite's breakdown strength. In the same way, there is the possibility of agglomeration in the sample due to higher filler loading and this leads to earlier breakdown.^[59,61] Si-Jiao Wang et al., revealed that the addition of modified Al₂O₃ nanoparticles to LDPE improves a variety of electrical

properties. Surface treated Al₂O₃ nanoparticles with vinyl trimethoxy silane are included in the polymer matrix and a thin-film sample with a thickness of 30 μm is stressed by the DC voltage at a rate of 500 V/s. According to the findings in Figure 7B,C, the nanoparticles in the polymer matrix form deep traps, limiting carrier mobility, and increasing resistivity. As a result, the surface-modified Al₂O₃ nanocomposite has a higher breakdown strength than untreated nanocomposite and neat polymer.^[47]

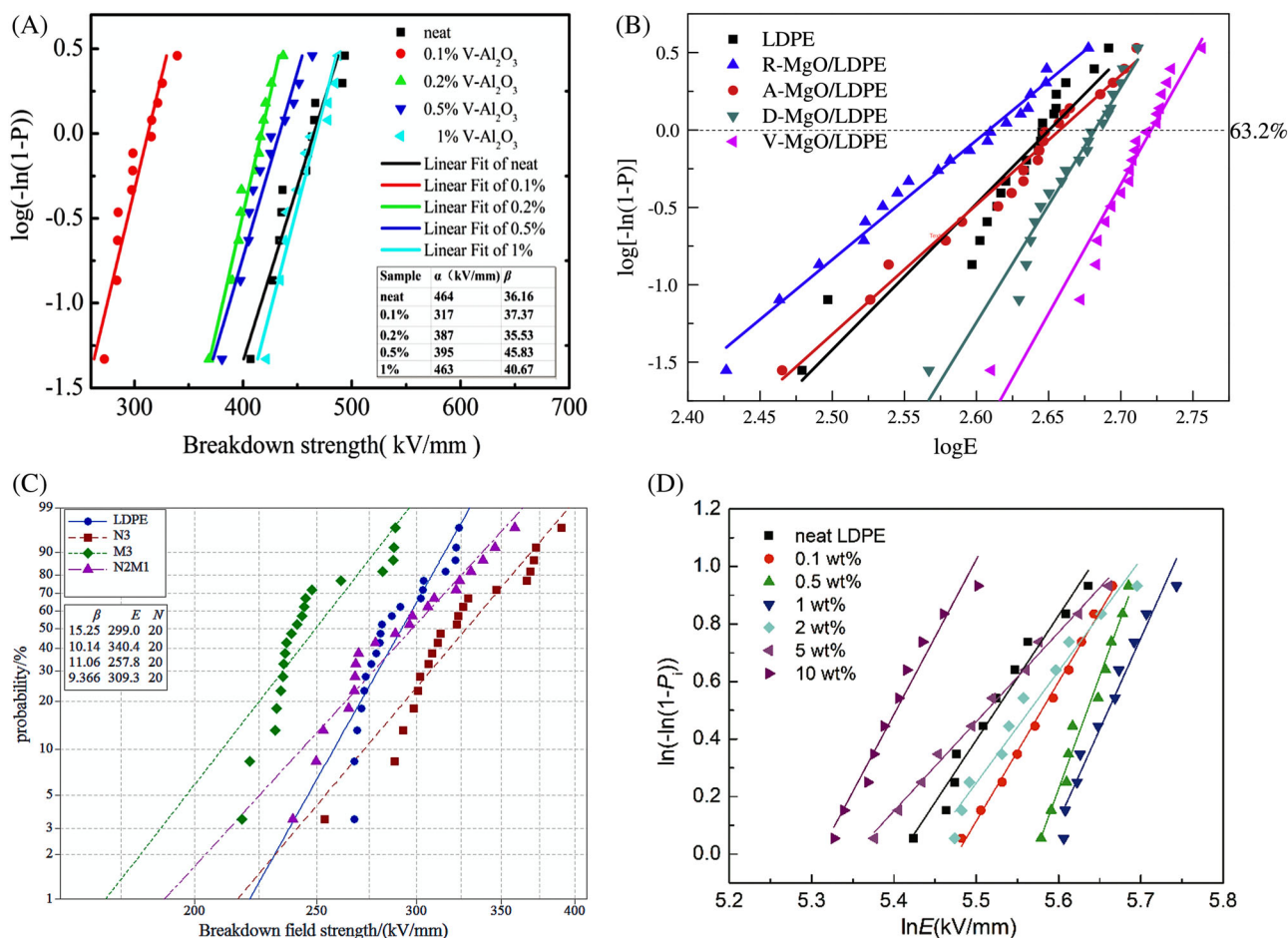


FIGURE 8 Breakdown characteristic of (A) LDPE/V-Al₂O₃,^[46] (B) LDPE/MgO,^[61] (C) LDPE/ZnO,^[47] and (D) LDPE/TiO₂ nanocomposite.^[48]

When compared to carbon black nanoparticles, Al₂O₃ nanocomposites have superior electrical properties and improve breakdown strength with filler loading, whereas carbon black nanoparticles have the opposite effect, as shown in Figure 7A. According to Siting Ju et al., reported that the inclusion of surface-modified MgO nanoparticles improves the electrical properties.

The nanoparticles' surfaces are modified with amino and vinyl functional groups and it is stressed at a rate of 2 V/s. According to the findings, the breakdown strength of pure LDPE is 442 kV/mm and increases with filler loading. At different temperatures, unmodified MgO has a slightly lower breakdown strength than pure LDPE, whereas functionalized nanofiller loading has a higher breakdown strength. The vinyl functionalized MgO nanocomposite achieves higher values (510 kV/mm), which are 15% higher than the neat LDPE.^[61] A small group of researchers recently discovered that adding micro-nano particles to a polymer matrix improves dielectric properties.^[45,47,62] Jun-Guo-Gao et al., reported that the properties of LDPE nanocomposites are altered

by micro, nano, and micro-nano ZnO particles. Before being incorporated into a polymer matrix, 1 μ m and 30 nm ZnO particles are surfaces treated with a silane coupling agent (KH 570). A test sample with a thickness of 200 μ m is stressed by the DC voltage at the rate of 1 kV/s. According to the findings, the addition of ZnO fillers increases the breakdown strength of the nanocomposite, indicating that the fillers introduce deep traps that capture charge carriers and limit charge accumulation. The breakdown strength of nano-ZnO and micro-nano cofilled composites increases by 13.7% and 3.4%, respectively.^[47] Surfaced treated ZnO/LDPE composite shows higher breakdown strength than CB/LDPE, Al₂O₃/LDPE, and MgO/LDPE.

It has also been reported that surface-modified TiO₂ changes the dielectric property and increases the breakdown strength of the polymer composite. The TiO₂ nanoparticles (30 nm) are surface treated with (KH560) coupling agent before being incorporated into the polymer matrix with varying filler concentrations (0.1, 0.5, 1, 2, and 5 wt%) and the sample stressed at a rate of

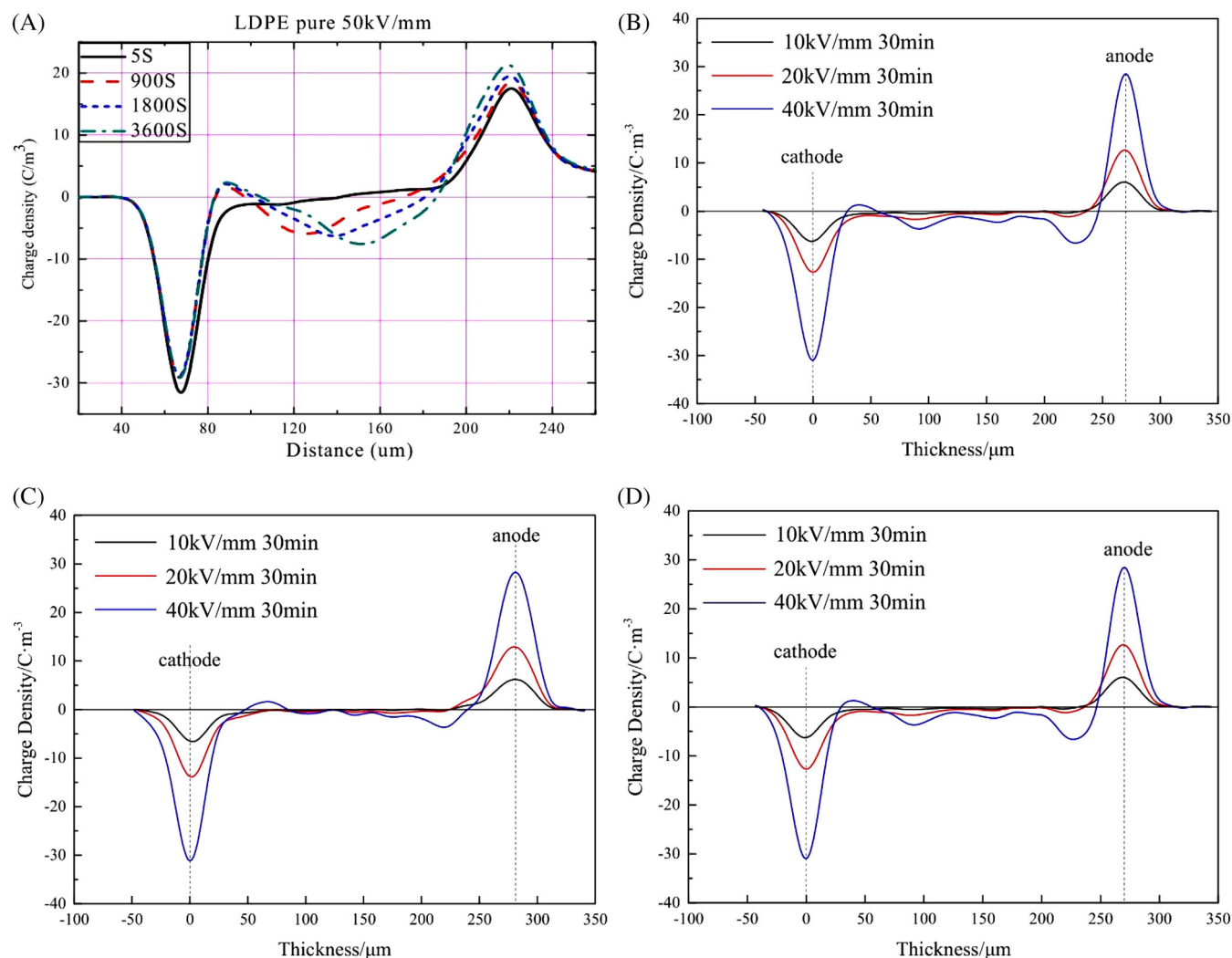


FIGURE 9 Space charge profile of (A) LDPE, (B) LDPE/ZnO, (C) LDPE/ZnO (micro), and (D) LDPE/ZnO (micro + nano).^[48]

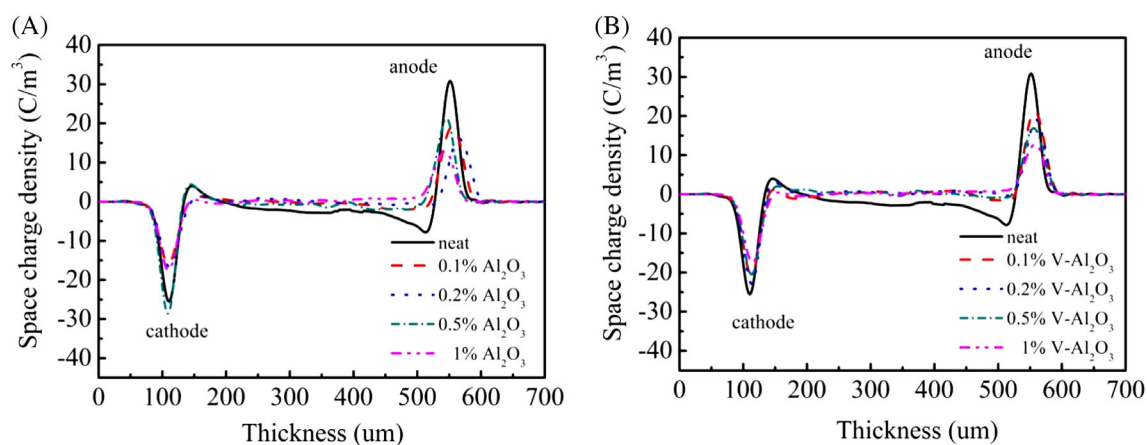


FIGURE 10 Space charge profile of (A) untreated Al₂O₃ (B) surface treated Al₂O₃ nanocomposite.^[47]

0.5 kV/s under DC voltage. It has been revealed that modifying traps, particularly deep traps, improves the dielectric breakdown of the composite. The effect of the

interface region changed the trap depth and density, resulting in a reduction in charge carrier mobility. At 1 wt%, TiO₂ nanocomposite enhances 19% higher

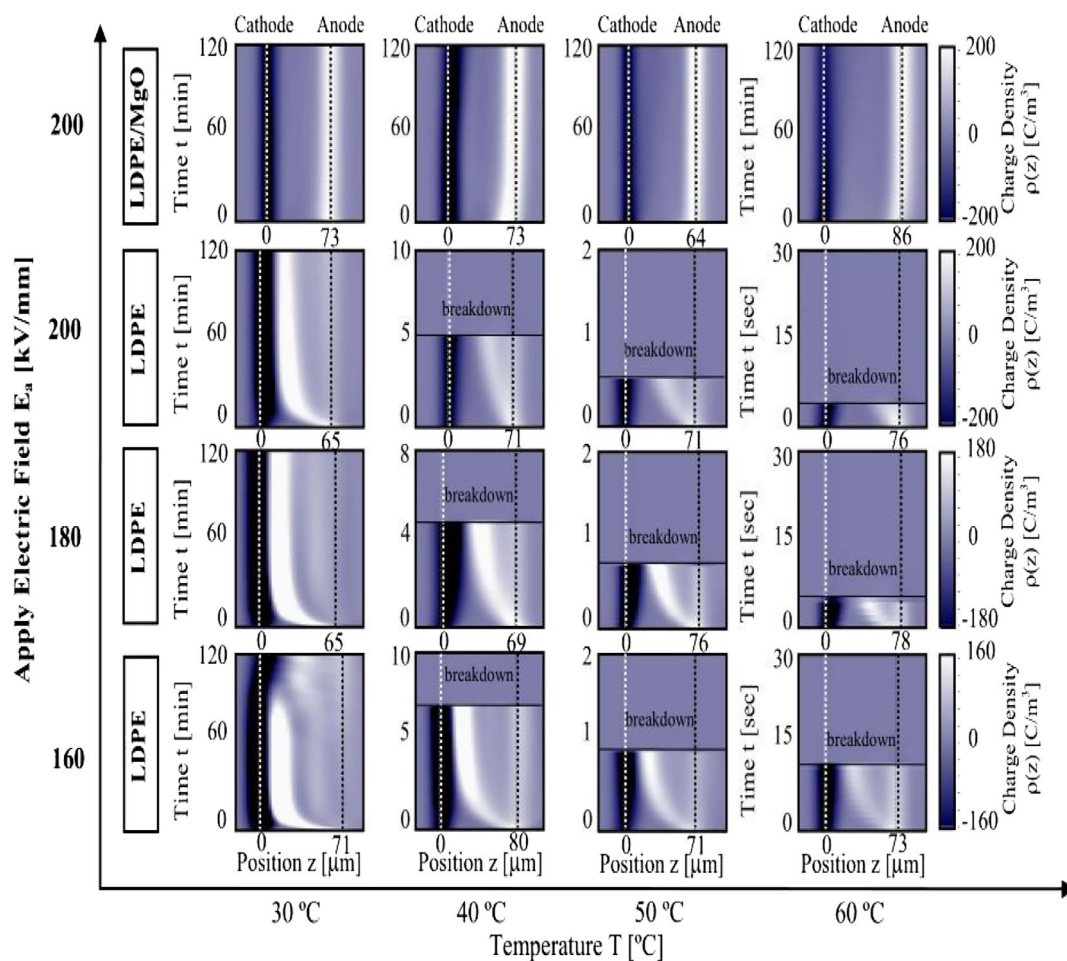


FIGURE 11 Space charge profile of LDPE and MgO/LDPE at various field and temperature.^[64]

breakdown strength than the neat polymer, but as filler loading increases, the breakdown strength begins to decrease, as shown in Figure 7D.^[49] According to the findings of various researchers on the LDPE nanocomposite under DC stress, TiO₂ has a higher breakdown strength among other fillers and the dielectric properties of the various nanocomposites are as shown in Figure 8.

2.2.8 | LDPE nanocomposite - space charge

According to the findings, Jun-Guo Gao revealed that the inclusion of micro, nano, and combined ZnO particles alters the properties of the LDPE nanocomposite. Before being added to the polymer matrix, the ZnO particles are surface treated with a silane coupling agent (KH 570), and a thin film sample with a thickness of 300 μm subjected for the space-charge distribution using the PEA method with various electric fields (10, 20, and 40 kV/mm).

As per Figure 9, there is no significant space charge accumulation under 10 kV/mm, but when it rises to

20 kV/mm, significant positive charges accumulate near the anode in neat LDPE and a spot of homo charges with a maximum charge density of 1.4 C/m³ arises near the cathode in nano-ZnO/LDPE. However, there is no significant space charge in the micro-ZnO/LDPE and micro-nano ZnO/LDPE composites. At 40 kV/mm, the positive space charge decreases in neat LDPE, while the negative space charge appears near the anode and the hetero charge accumulates near the cathode. However, there is no space accumulation within the micro-ZnO/LDPE or micro-nano ZnO/LDPE composites.^[48] According to Si-Jiao Wang et al., the inclusion of surface-treated alumina nanoparticles in the polymer matrix reduces the accumulation of space charges. The surface of the Al₂O₃ nanoparticles was treated with Vinyl Trimethoxy silane before being included in the LDPE polymer, and a thin sample with dimensions of 500 μm × 7 cm × 7 cm was used to measure the space charge profile using the PEA method. According to the profile shown in Figure 10, the space charge accumulation in LDPE/Al₂O₃ is less than that of pure LDP, and the suppression of space charge in V-Al₂O₃ is greater than that of the unmodified one.^[47]

TABLE 2 LDPE overview

Material	Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity (Ω cm)	Electric field (kV/mm)		
LDPE	Neat	AC	—	—	100–200	1	—	—	—	90 ^[44]		
				Cube-Cube	100	0.5	—	—	2.3×10^{17}	97.9 ^[44]		
	DC	—	Sphere-Sphere	100–300	1	—	—	—	299 ^[48]			
			Sphere-Sphere	300	2	Increase it lower frequency	Increase it lower frequency	2×10^{17}	400 ^[60]			
Composite		AC	ZnO MMT	Cylinder-Cylinder	30	0.5	Increase strongly with decreasing frequency	Nil	2.24×10^{16}	464 ^[47]		
				Cylinder-Cylinder	70	2	High permittivity at low frequency	High dielectric loss at low frequency	—	442 ^[62]		
				Sphere-Sphere	200	0.5	Stable	Stable	—	223 ^[49]		
				—	100–200	1	—	—	—	100 ^[45]		
				Self (Cube-Cube)	100	0.5	—	—	2.3×10^{17}	110 ^[46]		
				Sphere-Sphere	100–300	1	—	—	—	340 ^[48]		
				Sphere-Sphere	300	2	Increases with filler concentration	Increases with filler concentration	1.3×10^{17}	350 ^[60]		
		DC	Al ₂ O ₃ MgO TiO ₂	Cylindrical electrode	30	0.5	Decreases with filler concentration	Decreases with filler concentration	7.56×10^{17}	463 ^[47]		
				Cylinder electrode	70	2	Decreases with filler concentration	Decreases with filler concentration	—	510 ^[61]		
				Sphere-Sphere	200	0.5	Increases with filler concentration	Increases with filler concentration	—	265 ^[49]		

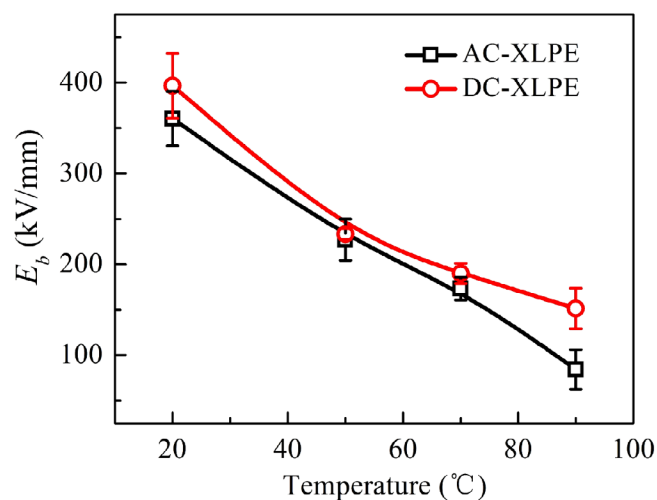


FIGURE 12 Breakdown characteristics of AC-XLPE and DC-XLPE.^[72]

The addition of montmorillonite (MMT) fillers to LDPE reduces the accumulation of space charges. MMT filler is surface treated with a silane coupling agent before being incorporated into the polymer matrix and the test sample is measured using a PEA system with a DC field of 50 kV/mm. According to the space charge profile, homo charges are injected into the polymer from both the anode and the cathode and move in bulk towards the opposite charge, with a large number of heterocharges accumulating near the anode ionization process. When compared to pure LDPE, the accumulation of space charge is reduced by 1 wt % MMT. However, at higher concentrations, a significant amount of heterocharges are observed, which are caused by impurity.^[46] Figure 11 depicts the space charge behavior of LDPE and composite materials in various applied fields (160, 180, and 200 kV/mm). A neat LDPE sample exhibits a pocket-like charge in all cases. At 30°C, none of the applied electric fields degrade, and the injected charges pass from anode to cathode. Only the neat LDPE, on the other hand, degrades at temperatures above 40°C.

As the packet-like charge moves towards the counter electrodes while gradually increasing its amount of charge, the electric field between the counter electrode and the position of the packet-like charge increases. Even at higher temperatures and voltages of 200 kV/mm, LDPE/MgO performs admirably. The MgO filler in the nanocomposite acts as a deep trap for the injected charges in strong electric fields and high temperatures.^[64] The researchers revealed that the MgO nanoparticles influence the space charge accumulation in a high electric field (200 kV/mm) with different filler concentration (0.2, 0.5, and 1 phr). There was no significant space charge observed in any of the filler loadings, but it is noticeable in a very small quantity of 0.2 phr when compared to the others.^[65] There was no space charge in MgO composite at 1 phr, according to the findings of the first two studies,

and the same was true in the high field at high temperature. Table 2 shows the overview of LDPE. The study of LDPE and its composites in the literature under various stresses are summarized as:

- Under AC stress LDPE exhibits low conduction current and the breakdown strength depends upon free electrons and energy gain.
- Under DC stress, the permittivity and dielectric loss of pure LDPE increase with a decrease in frequency, and also the conduction current increases with varying fields.
- MMT nanoparticles under AC stress reduce the charge carrier mobility and increase the resistivity of the LDPE composite.
- Whereas the ZnO nanofiller exhibits higher breakdown strength in both AC and DC studies. Under DC breakdown, ZnO composite exhibits 13.7% higher than pure LDPE and fails to suppress space charge accumulation.
- MgO nanofiller exhibits improved breakdown strength in both low and high fields. In low-stress fields, it improves breakdown strength by 15% over pure LDPE and there is no pocket observed.
- The TiO₂ nanocomposite outperforms pure LDPE and it is 19% higher breakdown strength. However, no evidence has been found in space charge studies so far.

Even though the dielectric properties of the composite materials get altered but the mechanical and thermal properties remain unchanged. To address the issue of LDPE, it has been replaced by a new polyethylene material called XLPE, which will be discussed further below.

2.3 | Cross-linking polyethylene

2.3.1 | Neat XLPE

To address the shortcomings of LDPE and HDPE, XLPE is obtained by cross-linking LDPE to improve thermal and chemical resistance, as well as mechanical properties, came into existence. Cross-linking polymer chains raise the operating temperature from 75 to 90°C. When compared to LDPE and HDPE, XLPE has superior insulation performance and thermal stability at higher temperatures (90°C). It has also been reported that XLPE remains stable at 130°C for 36 h even when the short circuit conductor reaches high temperatures (250°C). As a result, it is used as an insulation material in HVAC cables but fails in HVDC.^[66,67] However, the applications of XLPE in AC and DC power cables differ due to the different insulation performance requirements in the AC and DC fields. XLPE, on the other hand, XLPE has been used as an insulation material in HVDC cables for the

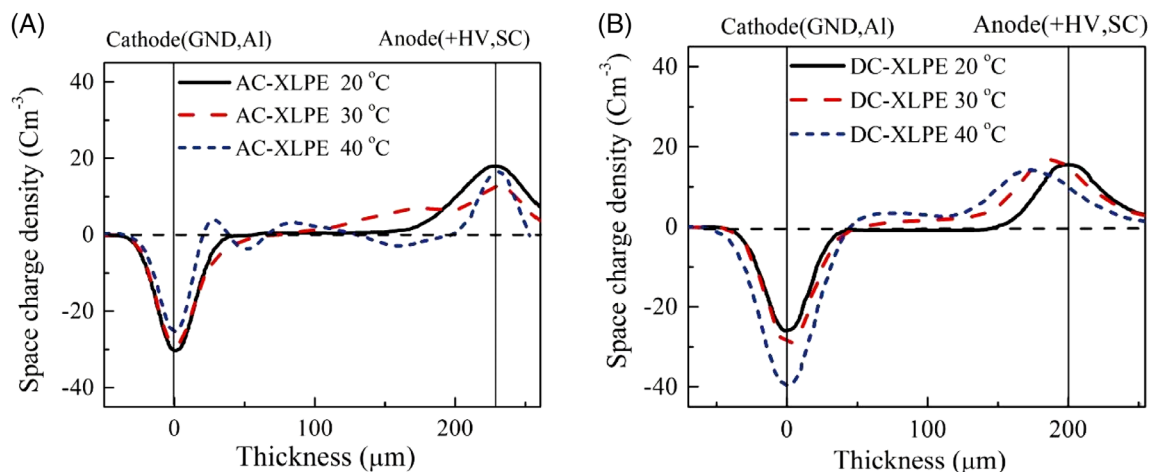


FIGURE 13 Space charge profile of XLPE (A) AC-XLPE and (B) DC-XLPE.^[72]

past 10 years as a result of extensive research.^[68,69] Despite its popularity, the XLPE has some disadvantages. When subjected to DC electric fields, space charge accumulates in the insulation, which is largely due to the influence of XLPE by-products. This accumulation causes severe field distortion and shortens the life of the cable.

2.3.2 | XLPE under AC stress

Zhipeng et al., reported on the impact of thickness on space charge behavior and dielectric properties of XLPE when stressed under AC voltage at a rate of 1 kV/s. From the observations, it depicts that the breakdown strength varies with thickness, as the thickness increases, the breakdown strength decreases. The breakdown always occurs at their weak points, which are caused by local defects. According to the free volume effect of solid dielectrics, increasing the thickness results in an increase in free volume, which increases the local field formation and thus reduces the breakdown field strength.^[70] The electrical properties of Silane cross-linked polyethylene(SXLPE) and DCP cross-linked polyethylene(DCXLPE) were reported. Furthermore, conduction is measured at applied electric stresses ranging from 0 to 40 MV/m at temperatures ranging from 30 to 90°C. According to the observations, the SXLPE has a lower conduction current than the XLPE and the higher value of conduction current is due to the bye products after cross-linking, which were a significant source of carriers. SXLPE has a higher breakdown strength than XLPE due to the absence of by-products and a lower conduction current.^[71]

2.3.3 | XLPE under DC stress

The performance of the XLPE is increased by its purity level. XLPE materials, which are used in AC and DC

cables, and the sample was stressed under DC voltage with a step voltage of 1 kV/s. It is clear from the observations that the breakdown strength decreases with increasing temperature. However, when compared to AC XLPE, DC XLPE has a higher breakdown strength, particularly at higher temperatures, as shown in Figure 12. As temperature rises, the DC conductivity of DC and AC XLPE rises exponentially but DC-XLPE has a higher electrical conductivity than AC-XLPE. The activation energies for DC-XLPE are 0.89, and 1.09 eV for AC-XLPE, respectively.^[72] It has been reported that the effect of thickness.

on the breakdown strength of XLPE under AC stress, the breakdown strength of XLPE decreases continuously with increasing thickness, whereas under DC stress, it increases first and then begins to decrease, resulting in a higher breakdown strength at 100 μm thickness. The free volume effect explains the difference in breakdown strength; as sample thickness increases, the volume of the sample also increases, and also it increases the local defect, which leads to an earlier breakdown in AC. In addition to the free volume effect, field distortion within the sample caused by space charge is another factor for insulation breakdown under DC.^[70]

2.3.4 | Space charge on neat XLPE

S Wang investigated the effect of space charge accumulation in dielectric materials based on their purity level (AC-XLPE and DC-XLPE). AC-XLPE and DC-XLPE space charge profiles are measured in a DC applied field of 30 kV/mm. The space charge profiles of AC-XLPE and DC-XLPE were observed after 20 min at 20, 30, and 40°C.

According to the space charge profile depicted in Figure 13, the injected positive charges in AC-XLPE move towards the bulk and reach approximately the middle position of the sample. In the case of DC-XLPE, it

migrates deeper and closer to the cathode. However, negative charge injection is only visible in AC-XLPE and not in DC-XLPE. When the temperature rises to 40°C, negative charges accumulate near the anode while positive charges accumulate near the cathode.^[72] Under different operating conditions, the cross-linking by-products have an effect on space charge accumulation in the XLPE cable. The effect of thermal treatment and residues on space charge accumulation in XLPE cables was reported by the author.^[73] During voltage ramping, the modified and reference XLPEs' threshold stress of space charge generation differed significantly. In general, when residues were present, heterocharged regions formed, and when the residues were removed by degassing, no charge was formed. According to his findings, even after 24 h of 70 kV stress, there is no evidence of space charge accumulation in the modified degassed sample. The researcher^[70] reported on the effect of thickness on space charge accumulation. Using pulse electro-acoustic methods, the space charge profile of various thicknesses is obtained. Dielectrics of varying thickness exhibit different space charge phenomena due to the multiple trap distribution characteristics that regulate charge injection, migration,

and accumulation. The XLPE trap distribution affects not only carrier migration but also space charge accumulation. There are fewer chemical defects when the thickness is less than 100 μm , resulting in a lower density of deep traps and a higher charge accumulation depth due to the low injection barrier. The accumulation of more charge within the XLPE bulk is depicted in Figure 14. As a result, the distortion rate of field strength is 20% greater than the applied field strength. Finally, space charge behavior dominates the breakdown process. When the sample thickness reaches 100 μm , the charge accumulation within the bulk material decreases, and the breakdown strength rises. For XLPE with a thickness greater than 100 μm , the distortion rate of field strength is less than 20%, indicating that the volume effect dominates the breakdown process and, as a result, the breakdown strength decreases.

2.3.5 | XLPE nanocomposites

It is still being researched to improve the polymer's dielectric and mechanical properties. It is well

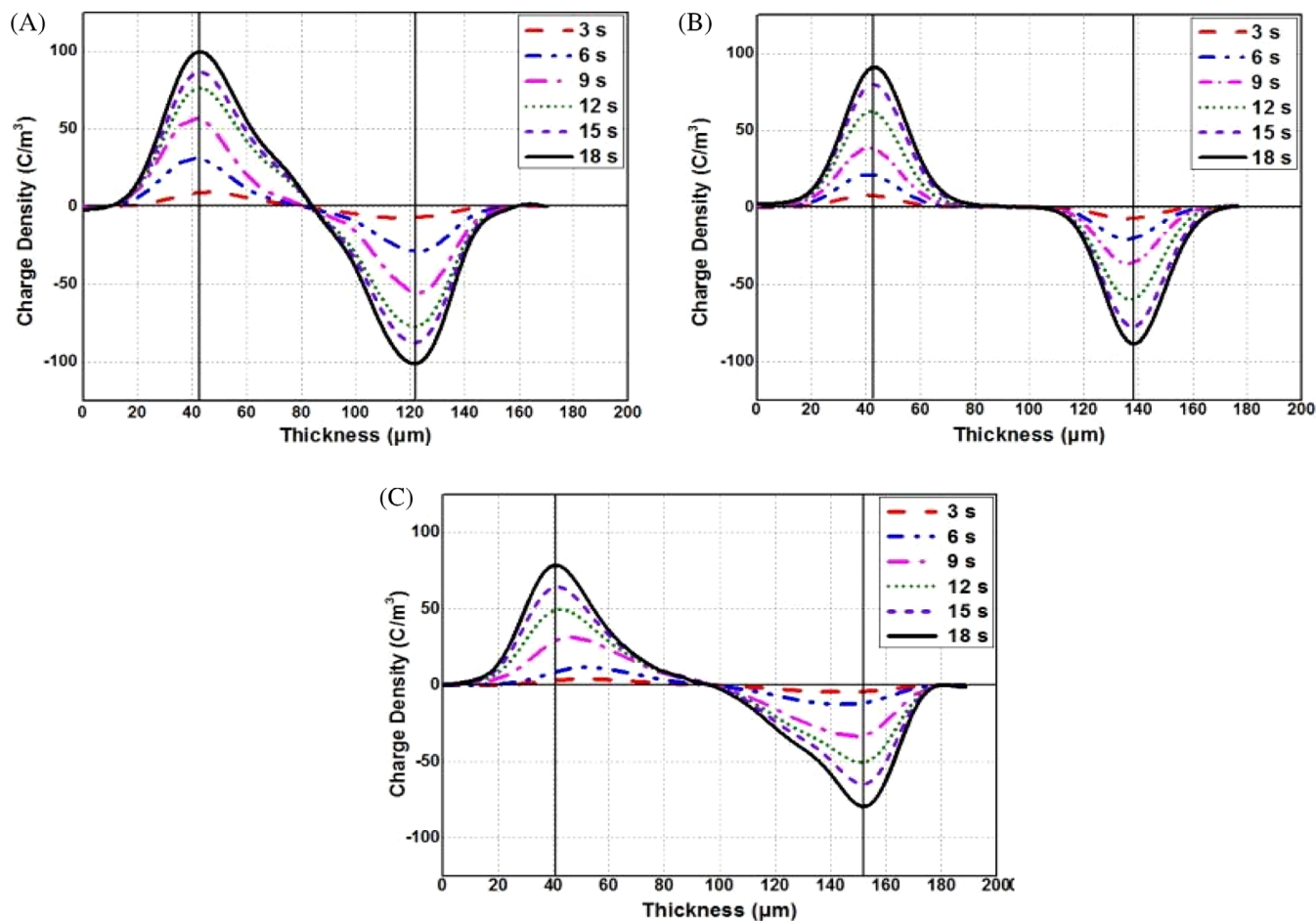


FIGURE 14 Space charge distribution with various thickness (A) 80 μm , (B) 100 μm , and (C) 120 μm .^[70]

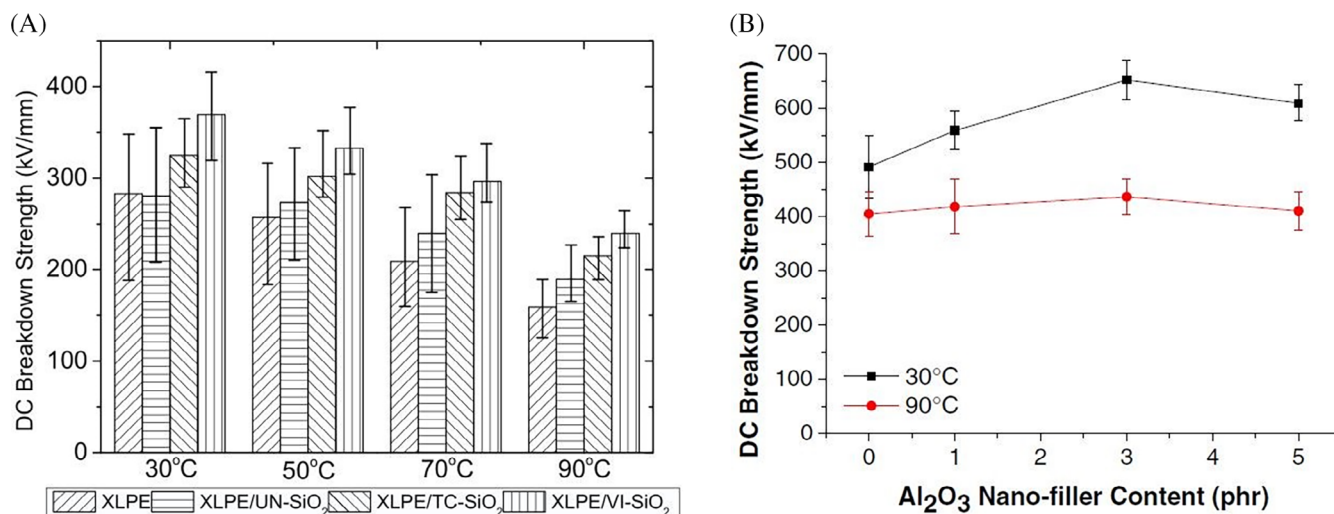


FIGURE 15 Break down strength profile of (A) XLPE/SiO₂^[77] and (B) XLPE/Al₂O₃ composite.^[78]

understood that the crystallinity of a material influences its breakdown strength. The nucleating agent and slower cooling rate control the higher crystallinity and smaller spherulite size. Only HDPE allows for the addition of a nucleating agent; LDPE and XLPE do not. The addition of an inorganic nanofiller to the polymer, on the other hand, acts as a nucleating agent in XLPE, reducing space charge accumulation while also improving the material's breakdown strength. The use of nanosized organic and inorganic fillers is becoming more popular. The interaction between the polymer matrix and nanosized fillers has a large surface area and hence it has a strong influence on them. The homogeneous dispersion of nanofillers has a greater influence on the properties of the nanocomposite and the surface modification prevents inorganic fillers from agglomerating and clustering in the polymer matrix.

2.3.6 | XLPE nanocomposite under AC stress

The addition of SiO₂ nanoparticles enhanced the dielectric properties of the XLPE material. The fumed silica with and without functionalization are incorporated in the polymer matrix using a suitable coupling agent. The addition of nanofillers improves AC breakdown strength with the functionalized nanofiller, whereas the sample with surface functionalized after heat treatment has a higher breakdown strength. When compared to the unfilled sample, the dc conductivity of the filled sample is reduced by 5 wt%, and the dc conductivity is further reduced after heat treatment. However, even after heat treatment, the conductivity of the unfilled remains unchanged.^[74]

2.3.7 | XLPE nanocomposite under DC stress

It is reported that the addition of TiO₂ nanofiller in the XLPE improves the dielectric property. The addition of unmodified and modified TiO₂ nanoparticles at different weight percentages alters the XLPE's properties (1, 3, and 5 wt%). According to the test results, the modified TiO₂ nanoparticle has better dispersion than the unmodified composite material. The volume of resistivity, as well as the breakdown, increases with filler concentration.^[75] It is also reported that the insulation performance of an XLPE nanocomposite via nanoparticle dispersion and distribution. TiO₂ nanoparticles with dimensions of (40 nm) are surface treated before being incorporated into the polymer matrix (KH560) and it is stressed at the voltage rate of 3 kV/s. The surface-treated nanoparticles in the polymer matrix are uniformly distributed, which helps to avoid agglomeration. The uniformity of dispersion decreases the DC conductivity of the nanocomposite material, whereas the neat sample has a higher DC conductivity. Similarly, surface treatment reduces DC conductivity in nanocomposite materials, which increases the breakdown strength of the material. As the filler content increases, the breakdown strength initially rises and then falls. When compared to neat XLPE, the XLPE/TiO₂-0.5 and XLPE/TiO₂ breakdown strengths are 13.5% and 7.8% higher, respectively.^[76]

The dielectric properties of XLPE are altered by the addition of surface-modified nanoparticles. Before being incorporated into the polymer matrix, the SiO₂ nanoparticles are surface treated with the coupling agents Titanate (TC) and vinyl silane. When compared to an untreated sample, surface-treated nanoparticles exhibit uniform dispersion in

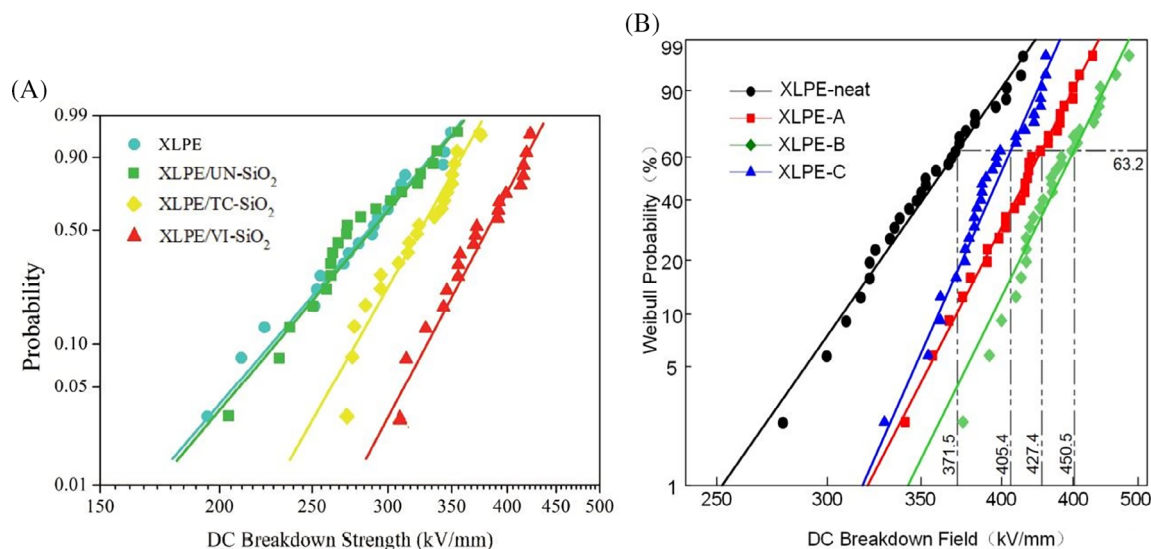


FIGURE 16 Breakdown strength profile of (A) XLPE/SiO₂^[77] and (B) XLPE with stabilizer.^[81]

the composite material. The coupling agent also modifies the interface between the XLPE matrix and the nanoparticle. As a result, the surface-modified nanocomposite outperforms the unmodified and neat XLPE in terms of breakdown strength. The breakdown strength of XLPETC- SiO₂ and XLPEVI- SiO₂ is 12.2% and 29.6% higher, respectively than that of neat XLPE. But the breakdown strength of neat XLPE and unmodified nanocomposite XLPE is similar, there are no significant differences in their properties.^[77] YongJun Park et al., reported on the improvement of XLPE properties through the inclusion of Al₂O₃ nanoparticles. According to the observations, the addition of nanofiller increases density, and thus the DC breakdown strength increases with increasing filler concentration up to 3 phr. Furthermore, as the content of Al₂O₃ nanofiller is increased until 1 phr, the volume resistivity of the specimen increases. From Figure 15B, the DC breakdown strength of a nanocomposite increases with filler concentration up to 3 phr and decreases above the filler concentration. The presence of filler in the polymer composite acts as a trap center, which increases the material's resistivity and consequently, it increases its breakdown strength. When the filler concentration exceeds 3 phr, the cross-linking in the polymer matrix weakens, resulting in a decrease in breakdown strength. Furthermore, when the temperature is raised to 90°C, the breakdown strength slightly decreases.^[78] In comparison to SiO₂ nanoparticles, Al₂O₃ nanoparticles in XLPE show higher breakdown strength under DC stress at 30 and 90°C, as shown in Figure 15.

The addition of MgO nanoparticles in the XLPE improved its electrical properties. Four different XLPE samples are considered: XLPE, XLPE with low content of cross-linking by-products and their composites with MgO nanoparticles. According to the findings, the volume of

resistivity is reduced in the XLPE sample due to the presence of by-products, but slightly higher in the XLPE composite sample. The addition of nanoparticles to pure XLPE increases its resistivity even more, while the volume of resistivity decreases as the field increases.^[79] Zhiyu yan et al., reported on the dielectric properties of the XLPE material altered by the inclusion of CB material. As the CB content of the XLPE/CB nanocomposite sample increases, so does its conductivity. This effect could be attributed to traps formed by the filling of nanoparticles, which reduce the migration rate. As the filler concentration increases, so does the trap concentration, and the traps encircle the majority of the carriers. The filler content also reduces space charge accumulation and field distortion. As a result, the electrical conductivity of composite material is greater than that of a pure material.^[80] Even though many researchers have explained how nanoparticles disperse well in polymers, it remains a challenge. The aggregation of nanoparticles will act as an impurity in the polymer matrix.^[76] It has been reported that using voltage stabilizers such as difluorobenzophenone, dihydroxybenzophenone, and Bisphenol (dimethylamino) improves the electrical and mechanical properties of polymers. It is discovered that the DC conductivity of all samples is of the same order of magnitude, indicating that the voltage stabilizer used in the experiment has no significant effect on the DC conductivity. The stabilizer, on the other hand increases the breakdown strength and makes it greater than that of the neat XLPE. And the breakdown value of the neat XLPE with other stabilizers is 371.5, 427.4, 450.5, and 405.4 kV/mm for XLPE, XLPE-A, XLPE-B, and XLPE-C, respectively.^[81] According to the breakdown strength profile shown in Figure 16, the XLPE with stabilizer has a higher breakdown strength than the polymer nanocomposite.

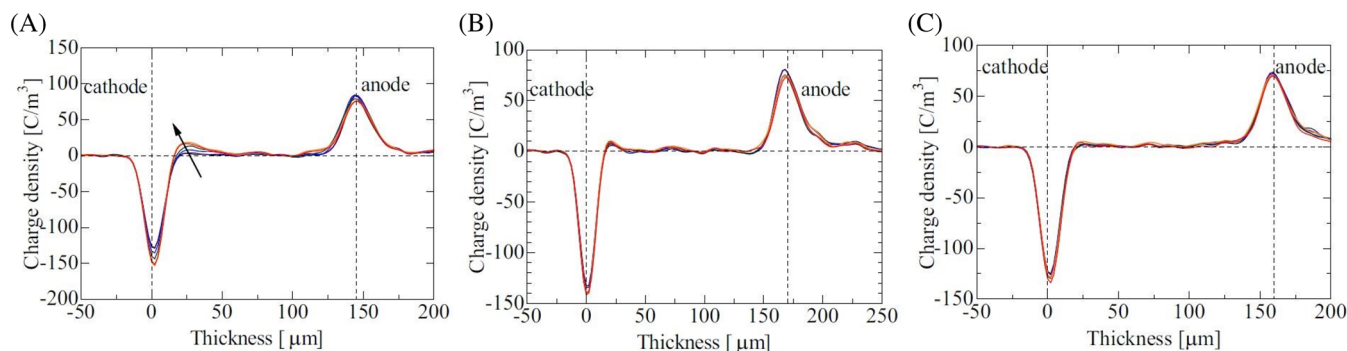


FIGURE 17 Space charge profile of XLPE/SiO₂ composite.^[74]

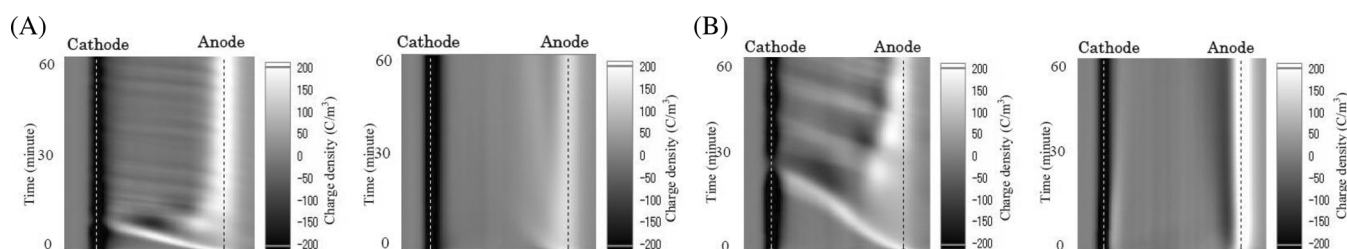


FIGURE 18 Space charge profile distribution of (A) XLPE(C), XLPE(C)/MgO, (B) XLPE(S), XLPE(S)/MgO composite.^[79]

2.3.8 | XLPE nanocomposite - space charge

The effect of surface-modified organic nanoparticles (SiO₂) on the properties of the XLPE material has been reported. The surface of the 50 nm nanoparticle is treated with aminopropyl triethoxysilane (KH550), propyltrimethoxysilane (KH560), and triethoxyvinylsilane (A-151) before it is incorporated into the polymer. Space charge was measured at room temperature using the pulse electroacoustic method with a field strength of 40 kV/mm. In the polymer, the surface-modified SiO₂ functions as a heterogeneous nucleation agent, reducing the melting point and shrinking the crystal size. The nanoparticle forms deep traps in the polymer matrix, binding the charge and reducing carrier mobility, thereby limiting space charge accumulation and migration.

Due to the heteropolar charges, the electric field in the sample is unevenly distributed, just like it is in pure XLPE. The addition of KH560-modified SiO₂ results in a crystal with a flawless structure and a lower rate of electrical distortion.^[81]

It is revealed that the surface-modified SiO₂ nanoparticles have an effect on the space charge accumulation of the nanocomposite material. The space charge profile demonstrates that positive and negative charges in the anode and cathode are injected into packets. Because positive charges decrease near the cathode while the electric field near the anode recovers, the distortion value will be high. In the case of XLPE/UN-SiO₂, the negative

charge collected by the shallow trap site could deep trap and recombine with the positive charge, resulting in a decrease in the amount of space charge and the homogeneity of the electric field distribution. Surface treated nanocomposite has less space charge accumulation than XLPE untreated nanocomposite and neat XLPE. The profile shows that the negative charges in XLPE/TC-SiO₂ move slowly and decay and that the negative space charge neutralizes the positive near the anode, lowering the local electric field. In the case of XLPE/VI-SiO₂, a higher concentration of traps was generated, capturing and blocking the injection and movement of packet-like charges.^[77]

It is also reported that the fumed silica nanoparticles on the accumulation of space charge in XLPE power cable insulation. The PEA method is used to assess three different samples: clean XLPE (H1), XLPE filled with 5% unfunctionalized SiO₂ (H2), and XLPE filled with 5 wt% functionalized SiO₂ (H3). According to Figure 17, in the neat XLPE, the homo charge and heterocharge are injected more easily. Whereas the inclusion of nanoparticles reduces charge packets, there is also very little accumulation of space charge in the surface-treated nanocomposite.^[74] Murata et al., discovered that incorporating MgO nanofiller.

into the polymer matrix improves space charge accumulation in the composite material. In this study, nanofillers are added to conventional XLPE(C) and XLPE with low crosslinking by-products XLPE(S). According to the results of the investigation, Figure 18 shows that the

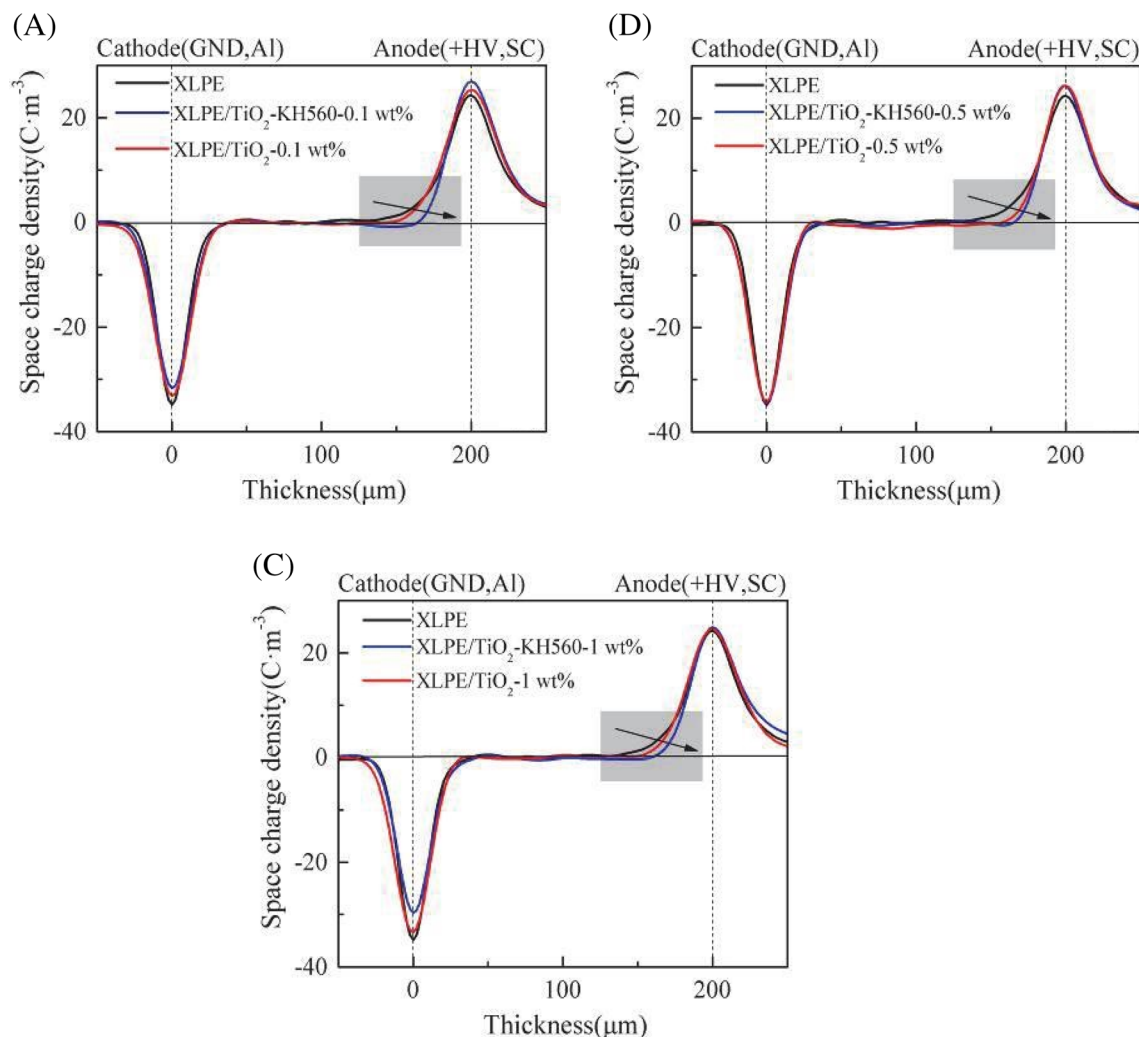


FIGURE 19 Space charge profile of XLPE TiO_2 at (A) 0.1 wt%, (B) 0.5 wt%, and (C) 1 wt% composite.^[74]

neat XLPE samples have packet-like charges that appear near the anode and shift towards the cathode. However, even at high fields, these packet-like charges were not observed in nanocomposites with MgO nanofiller.^[79] The same condition is replicated in LDPE with MgO nanofiller composite.^[64] Carbon black nanoparticles in the polymer matrix influence.

space charge accumulation. The space charge profile of a nanocomposite sample containing carbon black at various percentages (0.75, 1, and 1.5) phr is evaluated in this work using the PEA method at 40 kV/mm. According to the observations, there are several hetero charges at the anode and a small number of homo charges injected near the anode. At higher fields, the negative charge is ejected from the cathode, resulting in greater space charge accumulation.

When CB particles are added at 0.75 phr, the space-charge accumulation near the anode is greatly reduced, and this is effectively reduced the charger particle migration in the polymer. It is further reduced by introducing filler at a

rate of 1 phr in this situation, with really no charges near the electrode. When the filler concentration exceeds 1 phr, homocharges form between the two electrodes, causing the total charges inside the sample to rise.^[80] Shiang Wang reported on the space charge distribution of the XLPE/ TiO_2 nanocomposite material. The surface treatment of XLPE and XLPE/ TiO_2 nanocomposites with KH560 is investigated in this study using the PEA method at a field of 40 kV/mm. According to Figure 19, the presence of TiO_2 particles reduces homocharge injection and significantly reduces positive charge injection. Positive charges are suppressed even more by surface-modified nanoparticles, resulting in a higher material breakdown strength.^[76] Table 3 shows the overview of XLPE.

The study of XLPE and its composites in the literature under various stresses are summarized as:

- According to free volume theory, the breakdown strength of the neat XLPE is decreased as the thickness increases under the AC stress.

TABLE 3 XLPE overview

Material	Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity (Ω cm)	Electric field (kV/mm)
XLPE	Neat	AC	—	Sphere-Sphere	1000 μm	0.5 kV/s	2	10^{-5} – 10^{-3}	—	55 ^[74]
				Cylinder-Cylinder	100 μm	1 kV/s	—	—	—	140 ^[70]
	Composite	DC	—	Sphere-Sphere	17 μm	1 kV/s	—	—	—	400 ^[72]
				Cylinder-Cylinder	100 μm	1 kV/s	—	—	—	380 ^[70]
	Composite	AC	SiO ₂	Sphere-Sphere	1000 μm	0.5 kV/s	Higher at low frequency	Increase with filler	Increase with filler	52 ^[74]
		DC	Al ₂ O ₃	Sphere-Sphere	100 μm	1 kV/s	—	—	10 ¹⁵	600 ^[78]
			TiO ₂	Sphere-Sphere	150 μm	3 kV/s	—	—	—	350 ^[76]
			SiO ₂	Sphere-Sphere	1000 μm	0.5 kV/s	—	—	—	360 ^[76]
			Stabilizer	Sphere-Sphere	250 μm	0.5 kV/s	—	—	—	420 ^[81]

- On comparison to SXLPE and DCP-XLPE, SXLPE does not have any by-products, so there will be a lower conduction current and hence it has higher breakdown strength.
- Surface treated SiO₂ nanoparticles in the polymer matrix enhance the breakdown strength and further it was increased by the annealing processes under AC stress.
- Surface treated TiO₂ at 0.5 wt% shows 13.5% higher and at 1 wt% shows 78% higher than the neat XLPE.
- Surface treated SiO₂, XLPE-TC and XLPE-VI show high dielectric properties, which is 12.2% and 29.6% higher than the neat XLPE. Compare to Al₂O₃, SiO₂ shows higher break down strength, further the breakdown strength was enhanced surface treating with a silane agent.
- MgO nanoparticle improves the dielectric properties of the composites and also suppresses the space charge accumulation.
- Stabilizer such as dihydro-benzophenone and bisphenol shows uniform dispersion in the polymer matrix and hence it improves the dielectric properties.

As a result, the XLPE's breakdown strength improves. To summarize the overall report, XLPE is more prominent for HVDC cable insulation, but it still has disadvantages such as limitations in operating at higher temperatures and its thermosetting nature. Researchers are still working hard to develop new materials.

2.4 | Polypropylene (iPP)

2.4.1 | Neat iPP

Due to its recyclability and environmental concerns, the thermoplastic insulation material is attracting researchers to replace XLPE. iPP is a high-temperature-resistant thermoplastic polymer when compared to XLPE and LDPE, it has superior properties such as higher breakdown strength, higher resistivity, lower dielectric constant, and lower dielectric loss, as well as it does not have any by-products.^[83]

2.4.2 | iPP under AC stress

Hosier et al., reported on the properties of recyclable polymer materials. Conventional syndiotactic, isotactic, and iPP copolymers containing varying amounts of ethylene and butene monomers samples were stressed at a voltage rate of 50 V/s. The observations show that the overall breakdown strength of the iPP is higher (greater than 170 kV/mm) and also it possesses high thermal stability (above 160°C) than XLPE.^[84]

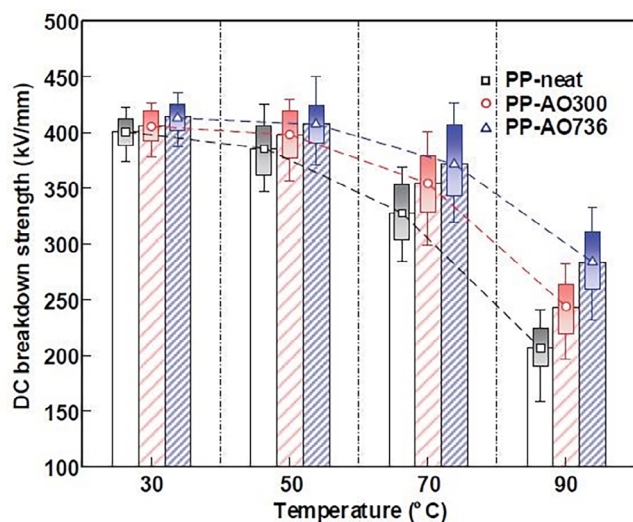


FIGURE 20 Breakdown characteristic of PP with antioxidant.^[84]

2.4.3 | iPP under DC stress

Xingyi Huang et al., reported on the potential of thermoplastic materials for HVDC cable insulation. Several studies on iPP, polypropylene block copolymer, and RCP are being conducted. A 150 μm thick thin film sample is stressed between the ball electrode at a voltage rate of 1 kV/s until it breaks down. According to the observations, the dielectric studies are carried out at frequencies ranging from 10^{-1} to 10^{-6} Hz, and the dielectric loss increases slightly as frequency decreases. It indicates that the leakage current has slightly increased. The breakdown strength of iPP is higher than that of block copolymers but lower than that of RCP. Furthermore, iPP has higher thermal stability and superior mechanical properties when compared to other thermoplastic materials.^[83]

The antioxidant in the polymer matrix also improves the properties of the polymer by reducing space charge accumulation and increasing the polymer's breakdown strength. According to reports, the two antioxidants improve iPP electrical properties. The polymer contains the antioxidants AO300 (6-tert-butyl-m-cresol) and AO736 (6-tertbutyl-0-cresol), and a test sample with a thickness of 70 μm is stressed at a rate of 1 kV/s under DC voltage. According to the results, the conductivity of the PP-AO300 and PP-AO736 is lower than that of the neat iPP. The conductivity of the PP-neat, PP-AO300 and PP-AO736 at 30°C is 6.46×10^{-16} , 4.29×10^{-16} , and 3.59×10^{-16} S/m, respectively. When the temperature was raised to 90°C, the conductivity of all three samples increased to 4.07×10^{-14} , 9.04×10^{-14} , and 9.91×10^{-14} , respectively. The breakdown strength of the antioxidant sample is greater than the breakdown strength of the neat

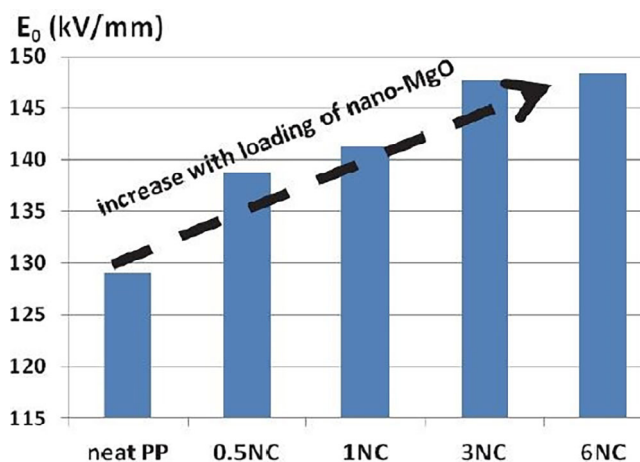


FIGURE 21 Breakdown characteristic profile of iPP/MgO composite under AC stress.^[86]

sample, and the breakdown value decreases from 401.0 to 206.4 kV/mm for AO300, 406.1–243.8 kV/mm for AO736, and 414.4 to 288.4 kV/mm for AO73 when the temperature is raised from 30 to 90°C, as shown in Figure 20. The decrease in value at higher temperatures is due to an increase in conductivity.^[85]

2.4.4 | iPP nanocomposite

Various research investigations report that inorganic fillers alter the dielectric properties of the polymer composite material. The inorganic nanofillers in the polymer matrix increase polymer crystallization by acting as a nucleation agent. In comparison to the unmodified sample, the inclusion of MgO nanofiller with and without surface modification in iPP raises the crystallization temperature and the breakdown strength.

2.4.5 | iPP nanocomposite under AC stress

Zhe Li et al., reported that the addition of MgO nanoparticles in iPP alters the electrical and thermal properties of the nanocomposites. MgO nanoparticles with dimensions of 30 nm are mixed with iPP with a melt index of 3.0 g/10 min and the test sample was stressed at a rate of 500 V/s under AC voltage (50 Hz). The breakdown strength of the polymer nanocomposite increases with filler loading under alternating current stress, as illustrated in Figure 21. This higher breakdown value is due to the filler's influence, which acts as a nucleating agent in the polymer composite and reduces the formation of the breakdown path.^[86] Thermally conductive aluminum nitride is functionalized with octyl-triethoxysilane before

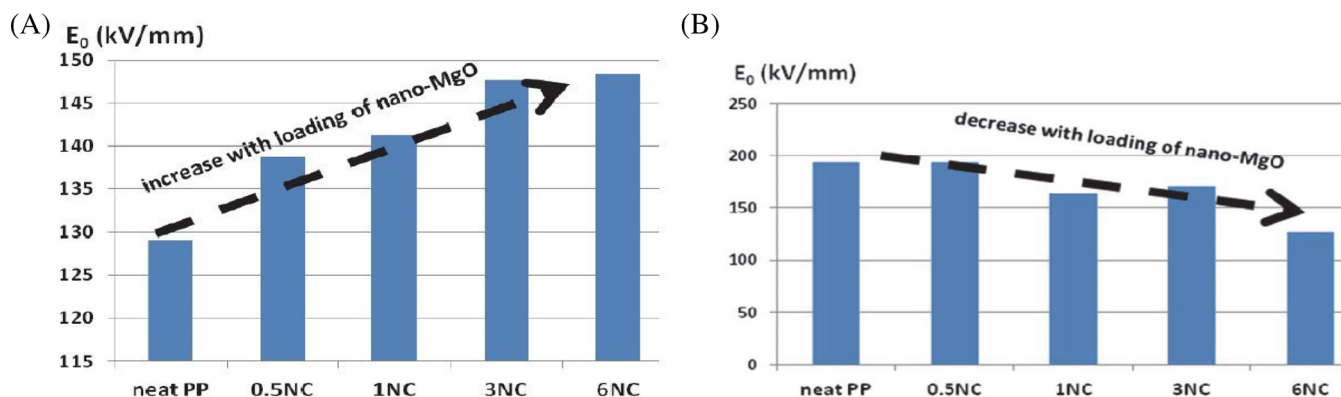


FIGURE 22 Breakdown characteristic profile of iPP/MgO composite under (A) AC and (B) DC stress.^[86]

being incorporated into a polymer matrix^[87] and the sample is stressed at a voltage rate of 0.5 kV/s. The conductivity of the nanocomposite increases with increasing filler loading at low and high fields. The lower breakdown strength value is caused by higher conductivity as well as a mismatch of permittivity and resistivity at the composite interface, which results in a severe local electric field and a lower breakdown value.

Even though it possesses a lower breakdown value, it is significantly higher than 380 kV/mm for SiO₂/LDPE composite and 390 kV/mm for SiO₂/XLPE composite.^[88]

Sendil et al., also reported on the addition of nanoparticles to iPP to improve the dielectric properties of the nanocomposites. In this work, various types of inorganic nanoparticles such as TiO₂, ZnO, and MgO are included in the virgin polymer at different weight percentages (1, 2, 3, 4, and 5 wt%) to create the thin film. Between the sphere and flat electrode, a thin with a thickness of 100 μ m is stressed. According to observations, the breakdown strength of the composite material increases with filler concentration up to 3 wt% and then begins to decline but TiO₂ shows higher breakdown strength than other composite materials.^[89]

2.4.6 | iPP nanocomposite stressed under high frequency

Sendil et al., also reported that the inclusion of nanoparticles changes the properties of iPP when it is stressed at different frequencies. In this work, nanoparticles such as TiO₂ and ZnO are included in the polymer matrix at varying weight percentages. A thin film with a thickness of 100 μ m is stressed between a sphere and a flat electrode using an alternating current voltage at different frequencies (10 and 20 kHz). According to the observations, the breakdown strength of the TiO₂ composite material increases with increasing

filler concentration up to 3 wt% and begins to decline at higher filler loading at both 10 and 20 kHz. It is 90% and 67% greater than the neat iPP stressed at high frequency, respectively. In contrast, the breakdown value of the ZnO/iPP nanocomposite did not change significantly. However, at 10 and 20 kHz, the breakdown profiles of both composites show similar variation. The reason for this is due to dipole re-orientation in the polymer matrix, which causes internal heating, which leads to earlier breakdown.^[90]

2.4.7 | iPP nanocomposite under DC stress

According to Zhe Li et al., the addition of MgO nanofiller changes the properties of iPP. In this study, a iPP nanocomposite (iPP/MgO) sample with different weight percentages of filler concentration (0.5, 1, 3, and 5 wt%) is stressed at a rate of 500 V/s under a DC voltage. The results show that as filler concentration increases, so does the DC breakdown strength of the iPP composite. Whereas the AC breakdown value increases with filler loading, the DC breakdown strength of nanocomposite is greater than the AC breakdown value as shown in Figure 22. The DC breakdown value of all composites is greater than 200 kV/mm, meeting the insulating strength requirement for DC power cable applications.^[86] Yao Zhou et al., also reported that the addition of aluminum nitride in the iPP alters the dielectric properties of the nanocomposites. In this study, aluminum nitride nanoparticles (50 nm) are surface treated with octyl-triethoxysilane before being added to a polymer matrix. The sample, with a thickness of 80 μ m, is sandwiched between two ball electrodes and stressed by a DC voltage at a rate of 1 kV/s. It depicts that the composite's conductivity increases with filler loading and hence, the breakdown strength of the composite decreases with an increase in filler concentration.^[87] The surface-treated

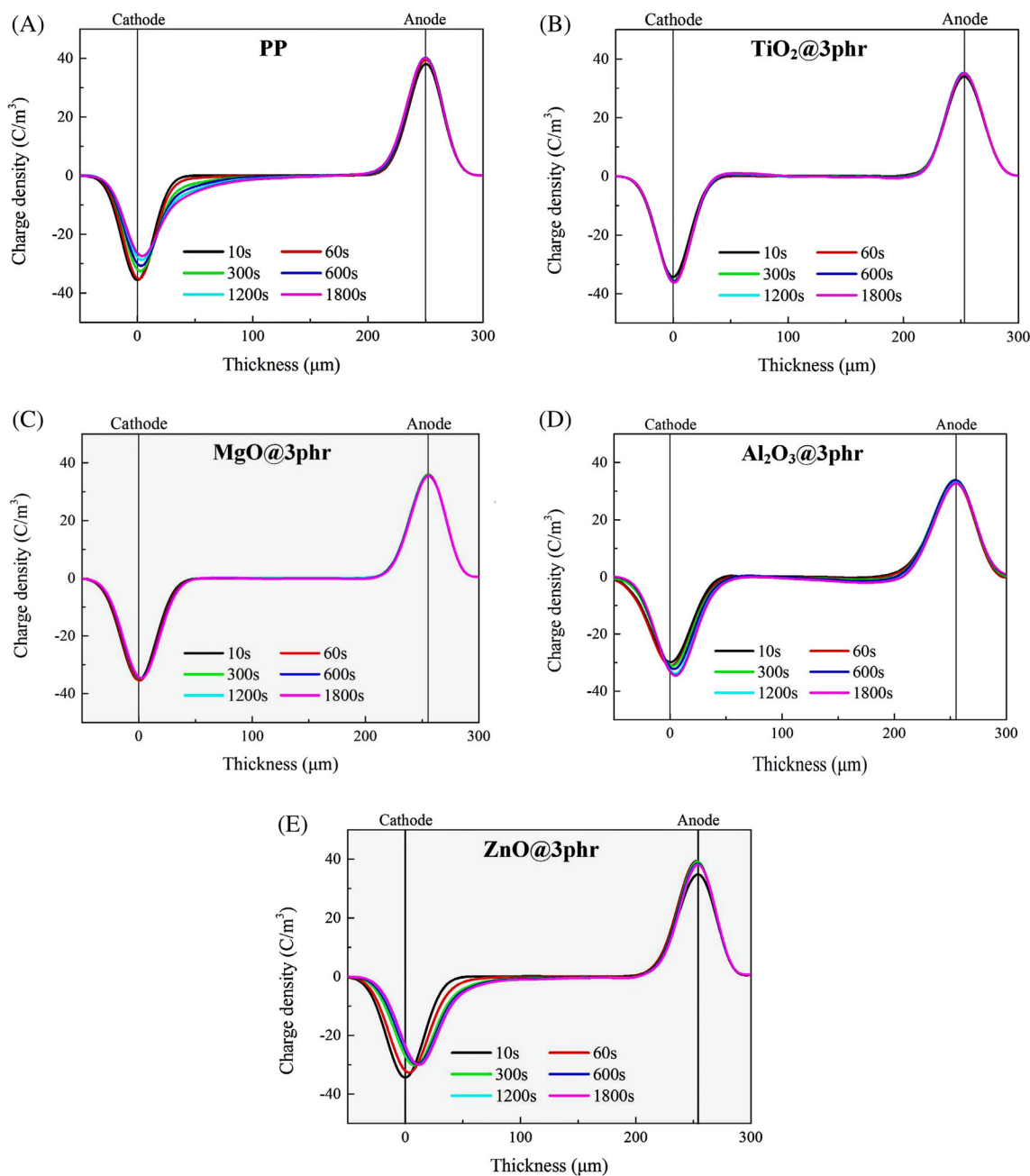


FIGURE 23 Space charge profile of (A) neat iPP and (B) TiO_2/iPP , (C) MgO/iPP , (D) $\text{Al}_2\text{O}_3/\text{iPP}$, (E) ZnO/iPP at 3 phr.^[92]

MgO nanoparticles improved the insulating properties of iPP. The MgO nanoparticle's surface was treated with aminopropyl trimethoxysilane before being added to a polymer matrix in this study. The breakdown strength.

of the surface-treated MgO nanocomposite increases with filler concentration and it demonstrates that at 1 wt %, the breakdown strength is 30.8% greater than pure iPP. The addition of MgO nanoparticles to the polymer acts as a nucleating agent, improving crystallization and thus increasing the breakdown strength of the nanocomposite material.^[89] Sendil et al., also reported that the inclusion of nanoparticles alters the properties of iPP. Nanoparticles

such as TiO_2 and ZnO are included in the polymer matrix with varying weight percentages in this work. According to the observations, the breakdown strength of the composite material increases with an increase in filler concentration up to 3 wt% and begins to decline at higher loading. At all weight percentages, the breakdown iPP nanocomposite outperforms the ZnO .^[88] Yao Zhou et al., also reported on the dielectric behavior of iPP with various nanoparticles (MgO , TiO_2 , ZnO , and Al_2O_3). The surface-treated inorganic nanoparticles with varying filler concentrations are introduced into a polymer composite. The dielectric constant of all fillers rises as the filler loading

TABLE 4 iPP overview

Material Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity (Ω.cm)	Electric field (kV/mm)
iPP	AC	—	Ball-Ball	70	0.5	—	—	—	173 ^[84]
		—	Sphere-Sphere	200	0.5	Increases with increase in temperature	Decreases initially and maintain stable almost	—	128 ^[86]
		—	Ball-Ball	80	0.5	Almost stable with increase in Frequency	Increase at a lower frequency and almost remain stable	5.5×10^{18}	189 ^[85]
		—	Sphere-Sphere	200	0.5	Increases with increase in temperature	Decreases initially and maintain stable almost	—	195 ^[86]
		—	Ball-Ball	80	0.5	Almost stable with increase in Frequency	Increase at a lower frequency and almost remain stable	5.5×10^{18}	512 ^[87]
	DC	—	Ball-Ball	120	1	Slightly decrease with increase in frequency	Maintain almost constant	—	389 ^[92]
		—	Sphere-Sphere	100	2	It increases with temperature after some point it start to decline	Decreases initially and maintain stable almost	—	209 ^[91]
		—	Sphere-Flat	70	1	—	—	—	400 ^[85]
		—	Ball-Ball	150	1	Remains almost constant	Increase at decrease in Frequency	—	340 ^[83]
		MgO	Sphere-Sphere	200	0.5	Increases with increase in filler concentration	Increases with increase in filler concentration	—	147 ^[84]
Composite	AC	AlN	Ball-Ball	80	0.5	Increases with increase in filler concentration	Increase at decrease in frequency	Decreases with filler concentration	166 ^[87]
		MgO	Sphere-Sphere	200	0.5	Increases with increase in filler concentration	Increases with increase in filler concentration	—	195 ^[86]
		TiO ₂	Ball-Ball	120	1	Increases with increase in filler concentration	Increases with increase in filler concentration	Decreases with increase in filler concentration	557 ^[92]
	DC	MgO						Increases with increase in filler concentration	503 ^[92]
		ZnO						Increases with increase in filler concentration	420 ^[92]
		Al ₂ O ₃						Increases with increase in filler concentration	458 ^[92]
								Increases with increase in filler concentration	

(Continues)

TABLE 4 (Continued)

Material Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity ($\Omega\cdot\text{cm}$)	Electric field (kV/mm)
Combined (AC + DC)		MgO	Sphere-Sphere	100	2	Decreases with frequency and weight percentage	Increases with increase in filler concentration	—	274 ^[91]
		AlN	Ball-Ball	80	0.5	Increases with increase in filler concentration	Increases with increase in filler concentration	—	460 ^[87]
		TiO ₂	Sphere-Flat	100	0.5	—	—	—	Increases with an increase in filler content decrease in presence of higher AC voltage ^[90]
		ZnO	Sphere-Flat	100	0.5	—	—	—	Increases with increase in filler content decreases in presence of higher AC voltage ^[90]

risers. This increase in the dielectric constant could be explained by interfacial polarization. For all composite materials, the dielectric loss increases but it remains lower than that of pure PP. Whereas the dielectric loss of TiO₂ nanocomposite is always greater than that of pure PP. Surface-modified nanoparticles in the polymer will act as deep traps at the interface, reducing the mobility of the charge carrier as well as the energy of the charge carrier and hence it resulting in an increase in breakdown strength. TiO₂ has the highest breakdown strength of 557 kV/mm among the four nanoparticles at 1 phr, followed by MgO, ZnO, and Al₂O₃ at 503, 420, and 458 kV/mm, with enhancements of 29%, 8%, and 18%, respectively.^[90]

2.4.8 | iPP nanocomposite under combined stress

When power is transmitted from the generating station, it is converted to DC and then back to AC at the receiving station. Power electronic circuits are used at both ends, some harmonics and other frequency components will be present in the DC voltage during transmission. As a result, the cable is stressed due to the presence of the harmonics and other high-frequency components present in the DC voltage. It is reported that when the sample is stressed under combined stress (AC + DC), the dielectric strength of the iPP nanocomposite gets altered. The TiO₂ and ZnO nanoparticles in polymer composite are stressed under combined stress (AC + DC), and the breakdown strength of the composite material decreases with increasing AC voltage. It is noted that in the combined stress, the high-frequency stress of 100 V decreases the DC breakdown voltage TiO₂/iPP, which is 50%–60% lower than the DC breakdown voltage, whereas for the ZnO composite it is 45%–90% lower.^[90]

2.4.9 | iPP nanocomposite - space charge

According to,^[91] the presence of MgO in the polymer composite alters the space charge profile. The space charge profile demonstrates that the accumulation of space charge is reduced when MgO nanofiller is introduced into the polymer matrix. As filler loading increases, the accumulation of space charge decreases, and this filler provides multiple scattering obstacles and trapping cite in the charge carrier path. As a result, the composite's carrier mobility and energy are reduced. Hetero charges and injected charges are absent in the polymer composite material when the filler loading is greater than 3 wt%. The presence of aluminum nitride also alters the space charge profile of the composite material.

According to the observations, there is no significant amount of space charge accumulation than the neat material, and it has uniform electric field distribution with an excellent profile when compared to commonly used plastic insulation such as LDPE and XLPE.^[87] It is clear that^[86] there is a packet of hetero charges observed near the cathode in the base material, and it increases with applied voltage but decreases when the MgO filler is introduced. There is no significant amount of accumulation at low filling concentrations, but homo charges are visible at higher concentrations of nanoparticles (6 wt%). This could be due to impurities introduced during the surface treatment of the nanoparticles. According to the data, homo charges are injected immediately after applying DC voltage, and the injection depth in the base sample increases with increasing polarization time. According to the space charge profile depicted in Figure 23, the addition of surface-modified inorganic nanoparticles (MgO, TiO₂, ZnO, Al₂O₃) reduces space accumulation. Apart from ZnO nanoparticles, the rest of them exhibit the most desirable

space charge accumulation at low filling concentrations of less than 3 phr, and there is still visible homo charge introduction in the ZnO nanocomposite, implying that space charge suppression is limited.^[92] Table 4 shows the overview of iPP.

The study of iPP and its composites in the literature under various stresses are summarized as:

- The antioxidant in the polymer reduces the conductivity in the polymer and increases the breakdown strength.
- Under AC stress, Aluminum nitride in the polymer increases the conductivity, and also a mismatch of permittivity and resistivity at the interface reduces the breakdown strength. The breakdown strength of the composite under AC and DC stress decreases with an increase in filler loading.
- Under high frequency the TiO₂ and ZnO show higher breakdown strength than the neat iPP at 10 kHz and 20 kHz.

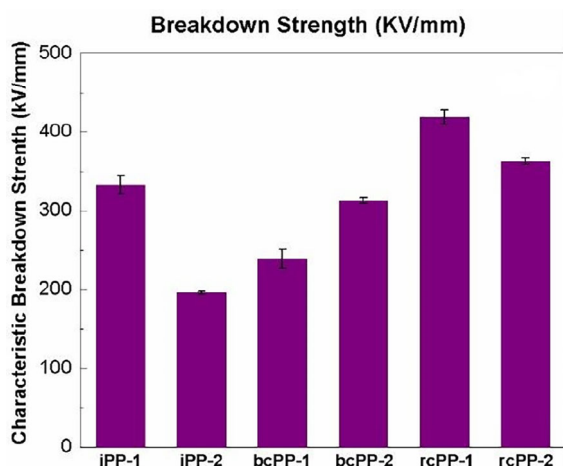


FIGURE 24 Breakdown strength of different thermoplastic polymeric material.^[83]

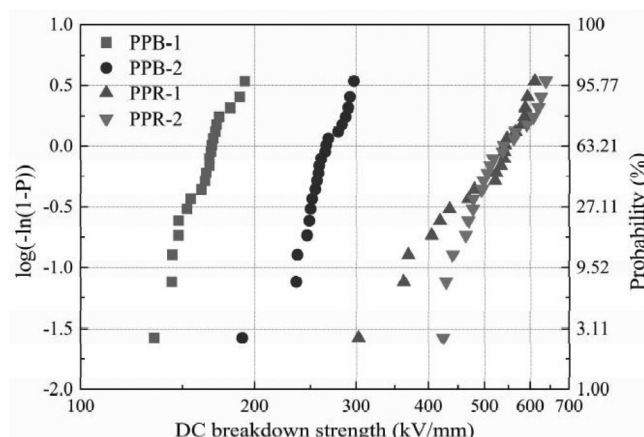


FIGURE 26 Breakdown characteristic of PP copolymers under DC stress.

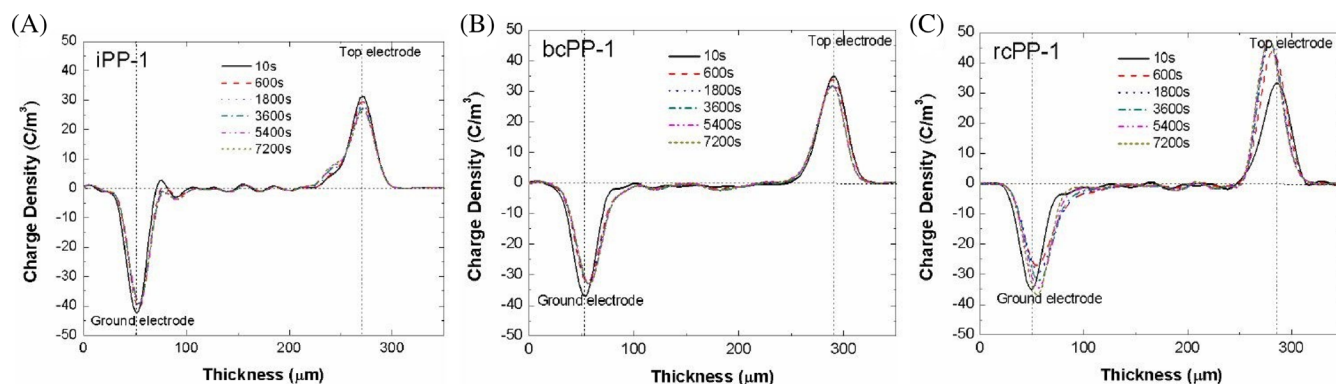


FIGURE 25 Space charge profile of PP copolymer.^[81]

TABLE 5 RCP overview

Material	Types	Stress	Fillers	Electrode configuration	Thickness (μm)	Rate of rise (kV/s)	Permittivity	Loss tangent	Resistivity ($\Omega \text{ cm}$)	Electric field (kV/mm)
RCP	Neat	DC	—	Ball-Ball	150	1	Remain constant with an increase in frequency	Increases at low frequency	—	420 ^[83]
				Ball-Ball	100	1	—	—	1×10^{18} – 1.5×10^{18}	540

- TiO_2 shows 90% and 67% higher than the neat iPP at high frequency but in contrast to ZnO/iPP did not significantly.
- Al_2O_3 in the polymer matrix shows higher breakdown strength than pure PP and also ZnO/iPP composite material, as well as it reduces space charge accumulation.
- The MgO nanoparticle in the polymer alters the electrical properties in both AC and DC stress. Under AC stress the breakdown strength increases with filler loading whereas in DC stress it declines. In addition, it suppresses the space charge accumulation.

From the overall data, it is clear that the nanoparticles in the polymer improve the dielectric properties of the nanocomposites but the drawback in the mechanical properties remains unchanged. Hence, scientists are conducting extensive research on thermoplastic materials that should overcome the aforementioned issues. To address the issues of iPP, it is replaced by RCP and which will be discussed below.

2.5 | Random copolymer

Polypropylene has drawbacks, such as brittleness at low temperatures and a lack of flexibility at room temperature, despite its high melting point, low dielectric loss, and high breakdown strength. To address the issue, iPP properties were improved by blending it with another monomer. However, the blending of monomers is a difficult task, and also it is facing a serious problem for compatibility and phase separation. Copolymerization is the most efficient method for creating new material and it avoids the problem as mentioned. To compensate for the problems associated with iPP, RCP are being used.

2.5.1 | Neat random copolymer

Xingyi Huang et al., reported on the thermoplastic iPP-based material for DC cable insulation. iPP, block copolymer, and RCP are all tested. Among all, iPP has the highest melting point, while RCP have the lowest.

RCP, on the other hand, has a higher breakdown strength than other polymers, as shown in Figure 24. Because of the low crystallinity, the breakdown strength and dielectric constant are higher. Similarly, due to the low crystallinity, there will be very little spherulite, resulting in a low accumulation space charge when compared to other polymer materials, as shown in Figure 25.^[83]

Pengfei Meng et al., revealed a comprehensive study on iPP copolymers. According to the findings, the volume of resistivity of RCP is greater than that of polypropylene block copolymers. And also homocharge injection occurs near the

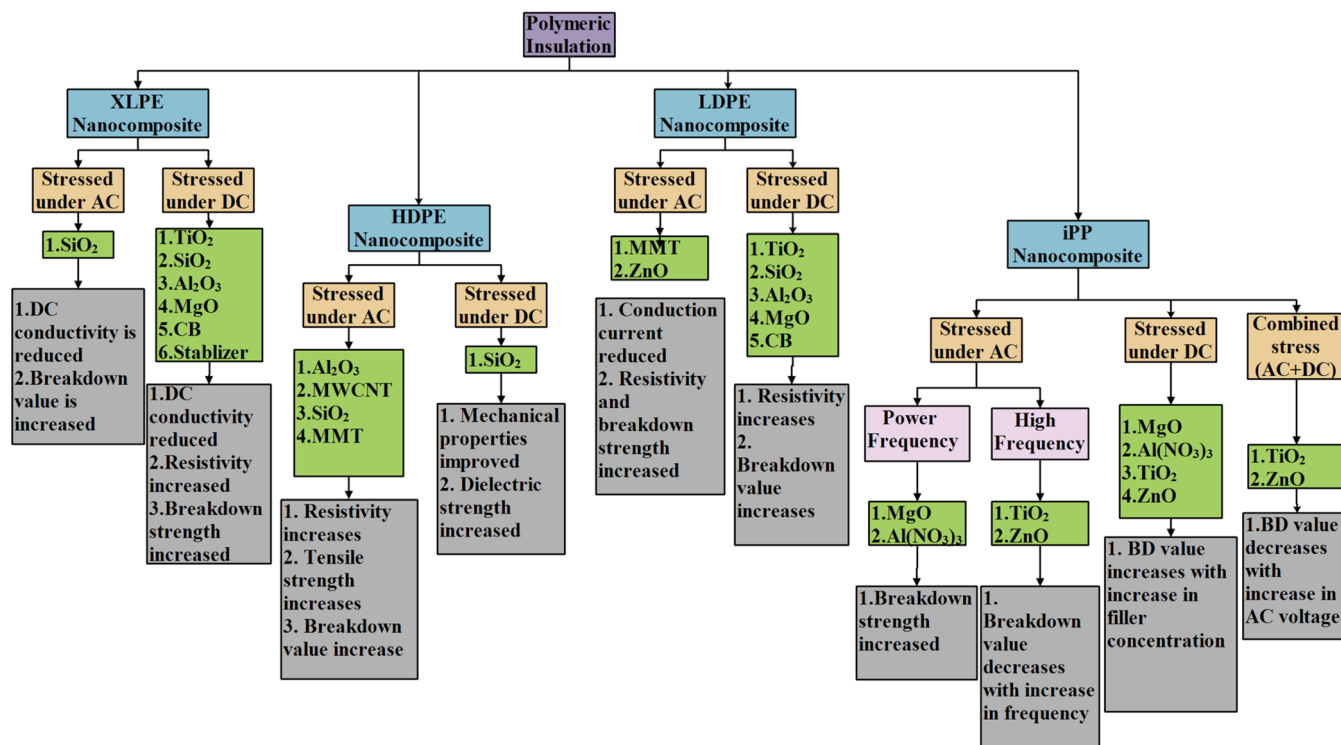


FIGURE 27 The role of polymeric composite under various stress conditions.

cathode in the PP copolymer, with little hetero space charge accumulation. According to Figure 26, as the polarization time increases, so does the depth of the injection, and the lower breakdown strength of the block copolymer is due to more injected homo charges, whereas the RCP has less.^[93] Table 5 shows the overview of RCP.

The study of RCP and its composites in the literature under various stresses are summarized as:

- The virgin RCP sample has a higher breakdown strength due to low crystallinity, small spherulite, and small space charge accumulation than another thermoplastic polymer.
- Furthermore, when compared to other polymers and composite materials, the homo charge injection is lower and also it has excellent mechanical properties. But there is no research evidence on the composite material.

From the overall reports of the various researchers, the role of nanoparticles under different stress condition are shown in Figure 27, this schematic view will helpful in selecting of nanoparticles for future research purpose.

3 | CONCLUSION

Overall reports from various researchers on various polymeric insulation materials are discussed in terms of

their electrical, mechanical, and thermal properties. Researchers have attempted to improve the properties of different polymeric materials through various modifications and manufacturing processes. The blending and copolymerization of new materials are being conducted to address concerns with polyethylene cables and to improve the insulation's electrical and mechanical qualities. Nanoparticles, on the other hand, play a significant role in the creation of improved base material properties. When compared to virgin material, the organic and inorganic-based fillers composite material have distinct properties. Nanoparticles form a strong bond with polymer even at low concentrations, influencing the properties of the nanocomposite material. The nanoparticles in the polymer agglomerate at higher concentrations, and it affects the performance of nanocomposites. Hence, to avoid this situation, the nanoparticles have been surface-treated with functionalizing chemicals, which will aid in uniform dispersion. When compared to the virgin material, the addition of nanoparticles, such as organic or inorganic nanoparticles, to polymer nanocomposite significantly changes the properties. Nanoparticles such as Al_2O_3 , MgO , ZnO , TiO_2 , carbon black, carbon nanotubes, and clay have been shown to improve electrical properties such as resistivity, breakdown strength, and also reduce space charge accumulation. Different fillers, however, have distinct properties in nanocomposites. It has been discovered that some fillers reduce the electrical

conductivity value to less than that of the neat material and that some filler loadings have negative effects on the underlying polymer, such as lowering the breakdown strength. The properties of the nanocomposite can be changed by choosing appropriate nanoparticles with low filler loading and fillers with surface treatment.

LDPE, HDPE, XLPE, and iPP are reviewed. In terms of dielectric and electrical properties, iPP and RCP outperform the remaining polymer materials. The breakdown strength of pure HDPE under AC stress is quite low whereas under DC stress it is high. The modified HDPE has low crystallinity and it possesses higher breakdown strength. Even though the addition of nanofiller improves the breakdown strength of the HDPE but it suffers from mechanical properties. To address the issue faced by HDPE it is replaced by the LDPE and also LDPE nanocomposites show excellent performance when compare to virgin material. Even though LDPE shows better performance, it has its own drawback. Later XLPE is widely used as an insulation material for HV applications. Even though the addition of nanofiller improves iPP performance in terms of properties, issues such as lack of flexibility and brittleness at low temperatures persist. Due to the non-recyclability and reuse of XLPE, many researchers work on the thermoplastic material to overcome the problem faced by the thermoset material.

Hence, RCP is a new material that addresses the limitations of iPP. While there have been a few research studies on RCP, there is still a lack of research on the effects of composite materials. RCP is a promising material for HVDC cable insulation. But there are very few research findings on the neat RCP by the researchers. The research investigation on space charge and PD in the neat RCP and RCP nanocomposites with various stresses is underway. And also to optimize the blending of iPP and RCP, and iPP /RCP with different nanocomposites with various weight percentages for HVDC applications.

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REFERENCES

- [1] N. M. Kirby, M. J. Luckett, L. Xu, W. Siepmann, HVDC transmission for large offshore windfarms, *Seventh International Conference on AC-DC Power Transmission* **2001**, p. 162. <https://doi.org/10.1049/cp:20010536>.
- [2] L. Xu, L. Yao, C. Sasse, *IEEE Trans. Power Syst.* **2007**, 22, 976. <https://doi.org/10.1109/tpwrs.2007.901306>
- [3] J. Lian, Y. Zhang, C. Ma, Y. Yang, E. Chaima, *Energ. Conver. Manage.* **2019**, 199, 112027.
- [4] J. W. Feltes, B. D. Gemmell, D. Retzmann, From Smart Grid to Super Grid: Solutions with HVDC and FACTS for grid access of renewable energy sources, *2011 IEEE Power and Energy Society General Meeting* **2011**, p. 1. <https://doi.org/10.1109/PES.2011.6039346>.
- [5] M. Jeroense, *IEEJ Trans. Electr. Electron. Eng.* **2010**, 5(4), 400.
- [6] D. Gielen, F. Boshell, D. Saygin, M. D. Bazilian, N. Wagner, R. Gorini, *Energ. Strat. Rev.* **2019**, 24, 38.
- [7] A. Cavallini, D. Fabiani, G. C. Montanari, *IEEE Electr. Insul. Magaz.* **2010**, 26(3), 7.
- [8] B. Mitra, B. Chowdhury, M. Manjrekar, *IET Renew. Power Gener.* **2018**, 12(13), 1563.
- [9] T. Worzyk, *Submarine power cables: design, installation, repair, environmental aspects*, Springer Science & Business Media, London and New York **2009**.
- [10] F. Miranda, P. G. Priaroggia, *IET* **1976**, 123, 229.
- [11] C. Arkell, A. Parsons, *IEEE Trans. Power Appar. Syst.* **1982**, 6, 1805.
- [12] X. Zhao, L. Pu, Z. Xu, G. Chen, W. Duan, H. Sun, Z. Ju, *High Volt.* **2020**, 5(5), 628.
- [13] R. Rendina, A. Gualano, M. Guarniere, *CIGRE B* **2008**, 1-104, 1.
- [14] S. Z. Ahmed Dabbak, H. A. Illias, B. C. Ang, N. A. Abdul Latiff, M. Z. H. Makmud, *Energies* **2018**, 11(6), 1448.
- [15] D. Wald, A. Smedberg, *IEEE Electr. Insul. Magaz.* **2008**, 24(5), 31.
- [16] A. Hedir, O. Jbara, M. Moudoud, F. Slimani, S. Rondot, F. Bel-labas, Behaviour of Cross-linked Polyethylene Used as High Voltage Power Cables Insulation Due to Environmental Constraints, *2018 International Conference on Electrical Sciences and Technologies in Maghreb (CISTEM)* **2018**, p. 1. <https://doi.org/10.1109/CISTEM.2018.8613596>.
- [17] C. Reed, *IEEE Electr. Insul. Magaz.* **2017**, 33(4), 22.
- [18] X. Liu, Q. Yu, M. Liu, Y. Li, L. Zhong, M. Fu, S. Hou, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1476.
- [19] Y. Liu, X. Cao, *IEEE Trans. Dielectr. Electr. Insul.* **2015**, 22(6), 3676.
- [20] G. Chen, M. Hao, Z. Xu, A. Vaughan, J. Cao, H. Wang, *CSEE J. Power Energy Syst.* **2015**, 1(2), 9.
- [21] D. He, X. Wang, H. Liu, Q. Li, G. Teyssedre, *IEEE Trans. Dielectr. Electr. Insul.* **2018**, 25(2), 541.
- [22] J. Li, Y. Gao, Y. Yuan, S. Huang, B. Du, Space Charge Accumulation in Polypropylene/Elastomer Blends with Different Elastomer Types, *2018 IEEE 2nd International Conference on Dielectrics (ICD)* **2018**, p. 1. <https://doi.org/10.1109/ICD.2018.8514662>.
- [23] M. E. Sepulveda-Garcia, J. M. Martinez-Tarifa, J. Sanz-Feito, *IEEE Trans. Dielectr. Electr. Insul.* **2013**, 20(6), 2222.
- [24] G. C. Montanari, P. Seri, X. Lei, H. Ye, Q. Zhuang, P. Morshuis, G. Stevens, A. Vaughan, *IEEE Electr. Insul. Magaz.* **2018**, 34(2), 24.
- [25] Y. Ouyang, M. Mauri, A. M. Pourrahimi, I. Östergren, A. Lund, T. Gkourmpis, O. Prieto, X. Xu, P. O. Hagstrand, C. Müller, *ACS Appl. Polym. Mater.* **2020**, 2(6), 2389.
- [26] Y. Zhou, S. Hu, C. Yuan, J. Hu, Q. Li, J. He, *CSEE J. Power Energy Syst.* **2020**.
- [27] Y. Zhou, J. He, J. Hu, B. Dang, *J. Appl. Polym. Sci.* **2016**, 133(1). <https://doi.org/10.17775/CSEEPES.2020.04740>.
- [28] U. T. Rahman, S. Amin, H. Shaikat, S. S. Haroon, I. A. Sajjad, M. Awais, *SN Appl. Sci.* **2019**, 1(10), 1. <https://doi.org/10.1002/app.42863>
- [29] E. David, M. Frechette, *IEEE Electr. Insul. Magaz.* **2013**, 29(6), 29.

- [30] C. Calebrese, L. Hui, L. S. Schadler, J. K. Nelson, *IEEE Trans. Dielectr. Electr. Insul.* **2011**, 18(4), 938.
- [31] T. Suzuki, S. Katakai, M. Kanaoka, Y. Sekii, DC characteristics of cable insulating materials, *Proceedings of the 3rd International Conference on Conduction and Breakdown in Solid Dielectrics* **1989**, p. 66. <https://doi.org/10.1109/ICSD.1989.69163>.
- [32] T. Tanaka, High performance HVDC polymer cable. Jicable'99 **1999**, 1989.
- [33] R. Asmatulu, W. S. Khan, R. J. Reddy, M. Ceylan, *Polym. Compos.* **2015**, 36(9), 1565.
- [34] G. Lessard, S. Péliou, H. St-Onge, M. Wertheimer, Comparison of high temperature breakdown behaviour in low and high density polyethylene, *Conference on Electrical Insulation & Dielectric Phenomena - Annual Report 1986* **1986**, p. 546. <https://doi.org/10.1109/CEIDP.1986.7726498>.
- [35] R. Ramkumar, C. P. Sugumaran, Investigation on dielectric properties of HDPE with alumina nano fillers, *2016 IEEE 7th Power India International Conference (PIICON)* **2016**, p. 1. <https://doi.org/10.1109/POWERI.2016.8077207>.
- [36] D. Li, L. Zhou, X. Wang, L. He, X. Yang, *Materials* **2019**, 12(11), 1746.
- [37] T. Suzuki, T. Niwa, S. Yoshida, T. Takahashi, M. Hatada, New insulating materials for HV DC cables, *Proceedings of the 3rd International Conference on Conduction and Breakdown in Solid Dielectrics* **1989**, p. 442. <https://doi.org/10.1109/ICSD.1989.69237>.
- [38] T. Niwa, M. Hatada, H. Miyata, T. Takahashi, *IEEE Trans. Electr. Insul.* **1993**, 28(1), 30.
- [39] T. Takahashi, H. Ohtsuka, H. Takehana, T. Niwa, *IEEE Trans. Power Appar. Syst.* **1985**, 8, 1945.
- [40] D. Muto, S. Kondo, K. Kaneko, Y. Suzuoki, T. Mizutani, T. Itaya, *IEEE* **1998**, 545. <https://doi.org/10.1109/ISEIM.1998.741801>.
- [41] T. Praveen, R. Sailaja, *Mater. Res. Bull.* **2016**, 83, 294.
- [42] O. E. Gouda, A. S. Haiba, *Measurement* **2019**, 134, 624.
- [43] K. Shah, R. Jain, V. Shrinet, A. Singh, D. Bharambe, *IEEE Trans. Dielectr. Electr. Insul.* **2009**, 16(3), 853.
- [44] V. Vahedy, *IEEE Electr. Insul. Magaz.* **2006**, 22(3), 13.
- [45] F. Tian, Q. Lei, X. Wang, Y. Wang, *IEEE Trans. Dielectr. Electr. Insul.* **2012**, 19(3), 763.
- [46] R. Liao, G. Bai, L. Yang, H. Cheng, Y. Yuan, J. Guan, *J. Nanomater.* **2013**, 2013, 1.
- [47] S. J. Wang, J. W. Zha, Y. H. Wu, L. Ren, Z. M. Dang, J. Wu, *IEEE Trans. Dielectr. Electr. Insul.* **2015**, 22(6), 3350.
- [48] J. G. Gao, X. Li, W. H. Yang, X. H. Zhang, *Crystals* **2019**, 9(9), 481.
- [49] L. S. WangW, *Materials* **2019**, 12(20), 3346.
- [50] T. Mizutani, H. Semi, K. Kaneko, *IEEE Trans. Dielectr. Electr. Insul.* **2000**, 7(4), 503.
- [51] G. Chen, T. Tay, A. Davies, Y. Tanaka, T. Takada, *IEEE Trans. Dielectr. Electr. Insul.* **2001**, 8(6), 867.
- [52] N. Amroun, M. Saidi, M. Bendaoud, R. Essolbi, T. Elallam, H. T. Giam, *Mater. Chem. Phys.* **1993**, 33(1–2), 168.
- [53] D. van der Born, I. A. Tsekmes, T. J. Person, S. J. Sutton, P. H. F. Morshuis, J. J. Smit, Evaluation of space charge accumulation processes in small size polymeric cable models, *2012 Annual Report Conference on Electrical Insulation and Dielectric Phenomena* **2012**, p. 669. <https://doi.org/10.1109/CEIDP.2012.6378869>.
- [54] N. Hussin, G. Chen, *IEEE Trans. Dielectr. Electr. Insul.* **2012**, 19(1), 126.
- [55] S. Wang, Q. Zhou, R. Liao, L. Xing, N. Wu, Q. Jiang, *Polymer* **2019**, 11(7), 1149.
- [56] T. Kato, R. Onozawa, H. Miyake, Y. Tanaka, T. Takada, *Electr. Eng. Jpn.* **2017**, 198(3), 19.
- [57] T. Maezawa, J. Taima, Y. Hayase, Space charge formation in LDPE/MgO Nano-composite under high electric field at high temperature, *2007 Annual Report - Conference on Electrical Insulation and Dielectric Phenomena* **2007**, p. 271. <https://doi.org/10.1109/CEIDP.2007.4451550>.
- [58] T. Takada, Y. Hayase, Y. Tanaka, T. Okamoto, *IEEE Trans. Dielectr. Electr. Insul.* **2008**, 15(1), 152.
- [59] J. Grunlan, F. Bloom, W. Gerberich, L. Francis, *J. Mater. Sci. Lett.* **2001**, 20(16), 1523.
- [60] S. Luo, S. Wang, Y. Tu, C. Wang, S. Qin, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1390.
- [61] Z. Yan, H. Zhao, B. Han, J. Yang, J. Chen, *e-Polymers* **2018**, 18(1), 49.
- [62] S. Ju, H. Zhang, M. Chen, C. Zhang, X. Chen, Z. Zhang, *Compos. A: Appl. Sci. Manuf.* **2014**, 66, 183.
- [63] M. Roy, J. K. Nelson, R. MacCrone, L. Schadler, *J. Mater. Sci.* **2007**, 42(11), 3789.
- [64] J. Yoshida, T. Maezawa, H. Miyake, Space charge accumulation and breakdown in LDPE and LDPE/MgO nano-composite under high dc stress at various temperatures, *2009 IEEE Conference on Electrical Insulation and Dielectric Phenomena* **2009**, p. 150. <https://doi.org/10.1109/CEIDP.2009.5377837>.
- [65] Y. Hayase, H. Aoyama, K. Matsui, Y. Tanaka, T. Takada, Y. Murata, *IEEJ Trans. Fundam. Mater.* **2006**, 126(11), 1084.
- [66] M. S. Khalil, On the use of doped polyethylene as an insulating material for HVDC cables, Vol. 2, *Conference Record of the 1996 IEEE International Symposium on Electrical Insulation* **1996**, p. 650. <https://doi.org/10.1109/ELINSL.1996.549429>.
- [67] Tanaka T, Greenwood A. *Advanced power cable technology. Volume I. Basic concepts and testing.* **1983**.
- [68] T. Goto, T. Yamazaki, T. Sugeta, K. Otake, I. Okajima, T. Sako, Recycling of silane cross-linked polyethylene for insulation of cables by supercritical alcohol, Vol. 3, *Proceedings of the 7th International Conference on Properties and Applications of Dielectric Materials (Cat. No.03CH37417)* **2003**, p. 1218. <https://doi.org/10.1109/ICPADM.2003.1218644>.
- [69] Z. Liu, J. Hao, R. Liao, J. Li, Z. Gao, Z. Liang, *IEEE Access* **2019**, 7, 165065.
- [70] Z. Ma, L. Yang, M. S. Bhutta, H. Bian, M. Z. Khan, *IEEE Access* **2020**, 8, 85552.
- [71] Y. Zhu, H. G. Yoon, K. S. Suh, *IEEE Trans. Dielectr. Electr. Insul.* **1999**, 6(2), 164.
- [72] S. Wang, P. Chen, H. Li, J. Li, Z. Chen, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1809.
- [73] M. Fu, G. Chen, L. A. Dissado, J. C. Fothergill, *IEEE Trans. Dielectr. Electr. Insul.* **2007**, 14(1), 53.
- [74] T. Tanaka, A. Bulinski, J. Castellon, M. Frechette, S. Gubanski, J. Kindersberger, G. Montanari, M. Nagao, P. Morshuis, Y. Tanaka, S. Pelissou, A. Vaughan, Y. Ohki, C. Reed, S. Sutton, S. Han, *IEEE Trans. Dielectr. Electr. Insul.* **2011**, 18(5), 1482.
- [75] Y. Wang, K. Xiao, C. Wang, L. Yang, F. Wang, *J. Nanomater.* **2016**, 1.

- [76] S. Wang, P. Chen, S. Yu, P. Zhang, J. Li, S. Li, *IEEE Trans. Dielectr. Electr. Insul.* **2018**, 25(6), 2349.
- [77] L. Zhang, Y. Zhou, X. Cui, Y. Sha, T. H. le, Q. Ye, J. Tian, *IEEE Trans. Dielectr. Electr. Insul.* **2014**, 21(4), 1554.
- [78] Y. J. Park, J. H. Kwon, J. Y. Sim, J. N. Hwang, C. W. Seo, J. H. Kim, K. J. Lim, *Jpn. J. Appl. Phys.* **2014**, 53(8S3), 08NL05.
- [79] Y. Murata, M. Goshowaki, C. Reddy, Investigation of space charge distribution and volume resistivity of XLPE/MgO nanocomposite material under DC voltage application, *2008 International Symposium on Electrical Insulating Materials (ISEIM 2008)* **2008**, p. 502. <https://doi.org/10.1109/ISEIM.2008.4664468>.
- [80] Z. Yan, B. Han, H. Zhao, J. Yang, C. Li, Space charge and conductivity characteristics of CB/XLPE nanocomposites, *Proceedings of 2014 International Symposium on Electrical Insulating Materials* **2014**, p. 30. <https://doi.org/10.1109/ISEIM.2014.6870712>.
- [81] B. Du, C. Han, J. Li, Z. Li, *IEEE Trans. Dielectr. Electr. Insul.* **2019**, 26(1), 34.
- [82] J. Kaihao, D. Donghe, X. Richang, W. Xiaoqiang, Y. Peijie, L. Xiufeng, Study on dielectric structure and space charge behavior of XLPE/SiO₂ nanocomposites, *2018 12th International Conference on the Properties and Applications of Dielectric Materials (ICPADM)* **2018**, p. 921. <https://doi.org/10.1109/ICPADM.2018.8401213>.
- [83] X. Huang, Y. Fan, J. Zhang, P. Jiang, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1446.
- [84] I. L. Hosier, S. Reaud, A. S. Vaughan, S. G. Swinger, Morphology, thermal, mechanical and electrical properties of propylene-based materials for cable applications, *Conference Record of the 2008 IEEE International Symposium on Electrical Insulation* **2008**, p. 502. <https://doi.org/10.1109/ELINSL.2008.4570382>.
- [85] B. Du, C. Liu, Z. Li, C. Han, Z. Hou, *IEEE Trans. Dielectr. Electr. Insul.* **2021**, 28(1), 124.
- [86] Z. Li, W. Cao, G. Sheng, X. Jiang, M. G. Danikas, *IEEE Trans. Dielectr. Electr. Insul.* **2016**, 23(3), 1812.
- [87] Y. Zhou, J. Hu, X. Chen, F. Yu, J. He, *IEEE Trans. Dielectr. Electr. Insul.* **2016**, 23(5), 2768.
- [88] S. Peng, J. He, J. Hu, X. Huang, P. Jiang, *IEEE Trans. Dielectr. Electr. Insul.* **2015**, 22(3), 1512.
- [89] A. S. Kumar, K. Elanseralathan, Assessment of Recyclable thermoplastic material for HVDC cable application, *2019 IEEE 4th International Conference on Condition Assessment Techniques in Electrical Systems (CATCON)* **2019**, p. 1. <https://doi.org/10.1109/CATCON47128.2019.CN0047>.
- [90] A. S. Kumar, D. Ashwini, S. Devnath, K. Elanseralathan, Effect of filler concentration on breakdown of Isotactic Polypropylene Nano-composites, *2019 IEEE 4th International Conference on Condition Assessment Techniques in Electrical Systems (CATCON)* **2019**, p. 1. <https://doi.org/10.1109/CATCON47128.2019.CN0097>.
- [91] W. Cao, Z. Li, G. Sheng, X. Jiang, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1430.
- [92] Y. Zhou, J. Hu, B. Dang, J. He, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24(3), 1380.
- [93] P. Meng, Y. Zhou, C. Yuan, Q. Li, J. Liu, H. Wang, J. Hu, J. He, *IEEE Trans. Dielectr. Electr. Insul.* **2019**, 26(3), 674.

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